Predicting Radiotherapy Sensitivity in Cancer Cell Lines from Epigenetic Data: A Machine Learning Approach



Marcelina Marjankowska

Supervised by Dr. Jamie Dean

University College London

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Abstract

Radiosensitivity is a major concern in the treatment of cancer. This study investigates whether machine learning models can predict radiosensitivity from multi-omics data, with a particular focus on global histone modification profiles, their integration with other modalities (RNA-seq, methylation, proteomics), and the transferability of models trained on cell lines to more clinically relevant systems. Key challenges were addressed using variational autoencoder–based data imputation, biology-informed dimensionality reduction, multi-head MLP architectures, and Bayesian neural networks.

RNA-seq was consistently the strongest modality (SVR/KRR, 0.30 R^2). Global histone modification data was a weak predictor (best 0.07 R^2) and typically degraded performance when combined with other modalities. Missing-value imputation improved both RNA-seq and methylation performance (to 0.30 and 0.22 R^2, respectively), suggesting that benefits extend beyond sample recovery to general noise reduction. Aggregating genes into pathway-level features with GSVA reduced dimensionality by $\sim$10–30$\times$ without loss of accuracy, and improved proteomics predictions (XGBoost, $R^2 = 0.281$).

Across integration strategies, multi-omic fusion did not surpass the best single-modality baseline. Although several combination approaches were tested, including multi-head MLPs, most yielded only moderate performance (0.22–0.26 R^2). Methylation and proteomics showed the most complementary signal, but RNA alone remained dominant. To better capture clinically relevant extremes, sample weighting improved predictions for highly sensitive and resistant samples, even when overall R^2 was unchanged. Finally, external validation on a more clinically relevant dataset showed partial transfer (AUROC = 0.69; Spearman's $\rho = 0.318$). This highlights the difficulty of translating predictive models from in vitro to in vivo settings and underscores the need for improved model systems as well as methods that explicitly address noise, imbalance, and domain shift.

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Chapter 1

Introduction

Cancer is one of the leading causes of death worldwide, and radiotherapy remains a cornerstone of treatment across many tumour types - approximately half of cancer patients undergo radiotherapy. However, tumours show highly variable responses to radiation, and there is currently no robust method for predicting how an individual tumour will respond. This dissertation investigates the use of machine learning methods to predict the sensitivity of cancer cells to radiotherapy based on molecular-level data.

1.1 Motivation and Clinical Importance

It has been noted that patients with the same tumour type can have very different responses to radiotherapy; some tumours are resistant, while others respond well. This variability is seen across cancers and is a key research problem [20]. For example, in head and neck cancer, some patients relapse after standard courses of radiotherapy while others achieve long-term control, even when their tumours appear clinically similar [20]. Such differences highlight the difficulty of predicting treatment outcomes. Aggressive treatment is linked to significant side effects, yet in some cases the tumour may be non-aggressive and a milder approach could suffice. At the same time, if a particular tumour is predicted to be resistant, alternative therapies could be considered earlier in the treatment plan. It is therefore important to choose the radiation dose carefully, balancing treatment efficacy with potential toxicity.

The Genomic Adjusted Radiation Dose (GARD) model [38] incorporates gene expression data into physical dose estimates. While early studies suggested that GARD could improve prediction of patient benefit from radiotherapy [39], the model has also faced significant criticism. At present, it is used in clinical trials and has

not been integrated into routine practice. This underscores both the promise and the current limitations of genomic approaches.

In recent years, increasingly detailed data on molecular, genomic, and epigenetic levels has become available for cancer cell lines through large-scale projects such as the Cancer Dependency Map (DepMap) [48]. Here, epigenetic refers to changes in gene activity (such as DNA methylation or histone modification) that do not alter the DNA sequence but can strongly influence cellular behaviour. While genomics has traditionally been central to identifying predictive biomarkers in cancer, recent large-scale functional screens by the DepMap have shown that mutations and copy number alterations alone may be insufficient to fully understand cancer biology [48]. This motivates exploration of more refined molecular features. The general approach of linking molecular profiles of cancer cell lines with treatment response has been successfully used in the context of drug screening. In particular, Seashore-Ludlow et al. [40] demonstrated how high-throughput profiling of cell lines against small molecules enabled discovery of drug response markers and mechanisms of action. Their work highlights the feasibility and value of integrating molecular features with treatment outcomes at scale. My work explores whether radiation sensitivity can similarly be predicted from underlying epigenetic and genomic characteristics.

Investigating the molecular-level and epigenetic data could reveal hidden differences between tumours and may help explain why cancer behaves so differently from one patient to another.

Terminology: Biological Data Categories

Genomic Data: DNA-level changes and inherited or somatic variants. This includes *mutations*, and *copy number alterations*.

Epigenetic Data: Regulatory mechanisms that influence gene expression without altering the DNA sequence. Examples include *DNA methylation* and *histone modification*.

Molecular Profiling Data: Molecular measurements of cellular activity, such as levels of RNA, proteins, or metabolites. This includes data from *transcriptomics (e.g., RNA-Seq)*, *proteomics*, and *metabolomics*.

1.2 Molecular-level Data for Radiosensitivity prediction

Large cancer cell line datasets enable ML application I make use of data from the Cancer Cell Line Encyclopedia (CCLE) [19], which includes genomic and epigenomic profiles of hundreds of cancer cell lines. Previous efforts using only clinical features or small-scale datasets offered limited predictive power. In contrast, the CCLE provides high-dimensional molecular data across hundreds of cancer cell lines, making it a valuable resource for training ML models.

This study focuses on in vitro cancer cell lines, which—while not fully replicating tumour microenvironments—have been shown to capture many biologically relevant features [40]. In particular, I focus on epigenetic layers: methylation, which can silence or activate gene regions, and histone modifications, which regulate chromatin accessibility and gene expression.

Radiosensitivity labels: AUC Radiation response is quantified using the RadioGx framework [34], which estimates dose–response curves for cell lines and summarizes them as area under the curve (AUC). AUC provides a global measure across multiple dose levels, in contrast to conventional single-dose metrics such as surviving fraction at 2 Gy (SF2). Prior work has shown that while SF2 remains widely used in large-scale radiogenomic analyses [35], AUC better reflects underlying biological processes, particularly in radioresistant cell types [34].

Patient-derived xenograft models of glioblastoma An important question in radiogenomic research is whether models trained on in vitro cancer cell lines can generalize to more biologically realistic systems. While cell lines are invaluable for generating large, standardized datasets, they lack the complexity of the tumour microenvironment and can accumulate genetic drift after extended culture [4]. In contrast, patient-derived xenograft (PDX) models—created by implanting human tumour tissue into immunocompromised mice—retain many of the molecular, histopathological, and treatment-response characteristics of the original patient tumours. A recently established panel of 96 glioblastoma PDXs has been shown to preserve clinically relevant therapy responses [49]. I use this dataset to validate my findings, providing a more translationally relevant benchmark for evaluating predictive models.

Epigenetic changes do not alter DNA sequence but affect gene activity Epigenetic changes regulate which genes are turned on or off without modifying the underlying DNA sequence. Common types include:

- *DNA methylation* – the addition of chemical tags to DNA, often affecting gene expression.
- *Histone modifications* – chemical changes to the proteins around which DNA is wrapped, influencing chromatin structure and accessibility.
- *Chromatin remodelling* – restructuring the packing of DNA to make certain regions more or less accessible.
- *MicroRNAs* – small non-coding RNAs that can block or reduce gene activity.

These modifications help organize nuclear architecture and regulate transcription, thus influencing processes such as cancer growth, progression, and metastasis.

Most existing treatment response prediction models rely on transcriptomics data such as RNA-seq, which provides a snapshot of gene expression in untreated cells. However, RNA-seq alone may miss mechanisms that are responsible for gene activation under radiotherapy. For example, if a gene is silenced by methylation, then following treatment it wouldn't be activated. RNA-seq data alone would not be able to capture that, whereas methylation data could reveal this hidden constraint. Epigenetic measurements therefore provide information that is complementary to transcriptomics, offering insights into how cells may respond to future perturbations.

For example, histone modifications have been linked to changes in the activity of many genes important for cancer development and progression [29]. In this dissertation, I focus on *global* histone modification profiles as ML features, which has been largely unexplored.

Predictive potential of Global Histone Modification data Histone modifications play a role in regulating chromatin structure and consequently gene accessibility. The human genome, which comprises over 20,000 genes, is tightly packed around histone proteins to fit within the nucleus. Chemical modifications to these histones—such as acetylation and methylation—can alter the compactness of chromatin, thus influencing gene expression.

Terminology: Histone Modifications

Histone: A type of protein around which DNA winds, helping package it into a compact structure. Core histones include H2A, H2B, H3, and H4.

Histone modification: A chemical change made to histone proteins. These changes can loosen or tighten how DNA is wrapped, which affects whether certain genes are turned on or off.

Global histone modification: The overall pattern or levels of specific histone modifications across the genome, rather than at individual sites.

While early studies primarily investigated histone modifications at individual gene promoters, more recent work has demonstrated that global patterns of histone modification—spanning broad chromatin regions—can carry prognostic value in cancer. In prostate cancer, specific global patterns of histone acetylation and methylation were shown to predict tumour recurrence, particularly in low-grade tumours. These patterns identified distinct molecular subtypes of disease and served as independent prognostic markers [42]. Several years later, the same authors identified patterns associated with survival outcomes in patients with lung and kidney cancer [41].

A related 2007 study [3] concluded that global histone modification patterns, particularly in H2A and H3, are significant prognostic markers in early-stage non–small-cell lung cancer. The authors suggested that such patterns could inform treatment decisions, especially in the context of epigenetic-targeting agents such as histone deacetylase inhibitors.

Despite these insights, the connection between global histone modifications and radiation response remains largely unexplored, representing an important direction for future research.

Multi-Omic integration It is important to use epigenetic data in combination with other modalities, particularly transcriptomics and proteomics. Hassanzadeh & Wang [21] demonstrated this in the context of cancer patient risk stratification, where integrating RNA, miRNA, and methylation data improved classification of high- versus low-risk patients in terms of overall survival. Their results support the idea that integrated molecular features can better capture treatment-relevant tumour biology. Moreover, they showed that the most informative modality depended on cancer type, which argues for using multi-modal data when prediction is performed across a wide range of cancers.

1.3 Previous Work

There has been relatively limited work on predicting radiosensitivity using machine learning applied to large genomic datasets. Many existing studies concentrate on smaller cohorts, identify candidate biomarkers, or address related problems such as drug response. Nonetheless, we can learn and draw from these adjacent results.

Radiosensitivity Prediction Several studies have attempted to predict radiosensitivity using machine learning, although many focus on limited sample sizes, single cancer types, or adjacent problems such as radiation toxicity or treatment response. Traditional approaches often rely on small, curated gene sets—ranging from 5 to 150 genes—and employing linear models, LASSO regression, or random forest classifiers. For instance, the widely referenced GARD model [38] uses a 10-gene signature with rank-based regression, while others explore immune, hypoxia, or methylation-related gene signatures using LASSO-based models [31, 8]. GARD however is based on the Radiosensitivity Index, which has is known to be unreliable. There was an attempt to use neural networks (NNs) and ResNet [27]; however, the results are unclear due to concerns regarding data leakage in testing. More recent work, such as Zeng et al. [55], proposes novel architectures like selectively connected neural networks driven by gene interaction patterns, demonstrating improved performance on small pan-cancer datasets. Another notable attempt involves transfer learning and variational autoencoders explored by a previous student [1]. In the context of nasopharyngeal carcinoma, a recent study evaluated 113 machine learning models and identified a combined glmBoost with Naïve Bayes approach as the most effective. This approach was used to develop a radiosensitivity score (NPC-RSS) based on 18 genes, which demonstrated strong predictive power in both public and in-house clinical datasets [31].

It is important to translate results to more clinically relevant models, as it is well established that findings from cell line studies often fail to translate directly to clinical settings [32]. In this work, we therefore test on mouse PDX models of glioblastoma. A related approach was explored by [33] in the context of drug sensitivity prediction, where few-shot learning was employed. However, their study had several limitations. Firstly, external validation was restricted to PDTC/PDX cohorts with very small sample sizes per drug (often ≤ 10), making reported gains highly sensitive to sampling. Secondly, performance varied considerably across tissues and drugs (e.g., Spearman's $\rho \approx 0.18$–0.50 in PDXs), raising concerns about robustness

and potential selection bias toward easier contexts.

One of the challenges in comparing results across studies is the use of varied response and evaluation measures. Some works [55, 27, 38] adopt the survival fraction at 2 Gy (SF2) index, which has been supported by integrative analyses of large-scale radiogenomic datasets [35]. However, SF2 captures response at a single dose point, whereas the area under the dose–response curve (AUC) incorporates information across a wider range of doses (e.g., 1–8 Gy). Notably, AUC has been shown to correlate with established biological processes related to radiation response and may provide a more informative measure, particularly for relatively radioresistant cell types [34].

Most existing studies, despite promising results, are constrained by limited molecular features or sample sizes. Comprehensive models leveraging large-scale multi-omics data (e.g., transcriptomics, epigenetics, proteomics) remain scarce. This dissertation aims to fill that gap by investigating the potential of deep learning methods to predict radiosensitivity using large-scale molecular datasets across diverse cancer types.

Other Machine Learning applications in cancer research Machine learning has also been applied to a wide range of problems in cancer research, including diagnosis, prognosis (e.g., stratifying patients by risk) [21], treatment selection [47], and cancer type classification [26]. It has further been used to identify biomarkers predictive of treatment response or disease progression—an important step for understanding cancer biology and informing the development of new therapies. A recent review highlights both progress and ongoing challenges in applying ML models for biomarker discovery in cancer prognosis and treatment response [9].

The closest and most extensively studied adjacent problem is drug response prediction [28, 36], which has even been used to guide large-scale drug screening efforts [40]. However, reported accuracies in some drug response studies can be misleading due to unrepresentative evaluation strategies. In particular, many works either (i) train on all drugs but withhold some data, or (ii) leave out only a small subset of drugs and attempt to predict their response. Both approaches tend to inflate performance because some cell lines are broadly sensitive to many treatments, and many drugs share highly correlated mechanisms of action. To avoid these pitfalls, in this work I employ a nested cross-validation design, ensuring that results are statistically robust and that the entire dataset is fairly evaluated.

Challenges Applying ML to omics data Applying machine learning to omics datasets presents several well-recognized challenges, many of which directly impact model performance and biological validity:

- *High dimensionality* — the number of features often far exceeds the number of samples, increasing the risk of overfitting.
- *Small sample sizes* — biological experiments are costly and time-consuming, limiting the availability of large, diverse datasets.
- *Noisy measurements* — both input features and response variables may contain substantial technical or biological noise.
- *Modality fusion* — integrating heterogeneous data types requires methods that can handle differences in scale, sparsity, and information content.
- *Missing values* — gaps may occur at the feature level or at the sample level, especially when combining multiple modalities.
- *Explainability and interpretability* — many complex models, particularly deep learning architectures, have poor biological interpretability, limiting their acceptance in translational research. A recent review highlights the need for explainable and transparent models in multi-omics applications [52].
- *Transferability* — models trained on large datasets collected under controlled laboratory conditions may not generalize to real-world or clinical settings.

Tackling high-dimensionality of multi-omics data The ML approaches in cancer research using multi-omics concentrate on tackling the high-dimensionality of data while being wary of relatively small number of samples. There is an emphasis on incorporating biological knowledge to drive model design, as noted in the review of DL models for cancer research by Wysocka et. al. [52]. Here are some of the most common approaches:

- *Sparsifying NN connections in a Biology-informed way.* Sparsely connected Deep Neural Networks are designed with connections defined by biological relations. One example being the P-NET where genes are arranged into pathways [13]. Another example is the ANN with Selective Connection based on Gene Patterns where first layer is based on gene interaction information [55].

Using NN helps models use a wider range of features and discover complex feature dependencies. However, NN will naturally overfit in the high-dimention small dataset setting, with the sparse connections helping to mitigate this phenomenom. The number of parameters however still remains large ($\sim$ 35.000 in the case of P-NET).

- *Graph Convolutional Networks* Another way of arranging genes into networks is by leveraging GNN or GCNs. They have been used for cancer type classification [37], and survival prediction [15]. Their strength is in being able to combine heterogeneous data types and leveraging domain knowledge such as protein-protein interaction network or pathways. However, they often either require prior biological knowledge to define connection, or the number of edges scales quadratically with the number of features.

- *Transfer Learning and multi-task learning* There are some much larger drug-response datasets [25, 40]. Because of that, transfer learning could be used to learn on these drug-response data and only then adapted to radiotherapy-response data. This has been explored by a previous student, but proved to be unsuccessful [1]. Transfer learning can also be used for moving models from cell line data to more realistic settings, which was done in the context of drug response [33, 43, 22].

- *Low-dimensional representation* can be obtained using encoder–decoder architectures such as VAEs [44, 23], which are particularly well suited to handling sparse data. More commonly, classical methods such as PCA or predefined subsets of informative genes are used, with the resulting representation then fed into the ML model [7]. Subset selection also aids interpretability, but it is often unstable in high-dimensional settings—small data perturbations can yield very different selected features [52].

- *Semi-supervised learning* Unlabeled multi-omics data can be used for unsupervised pretraining to improve model robustness when labeled samples are scarce [43]. For example, Hassanzadeh & Wang (2021) used a deep belief network pretrained on RNA-seq, miRNA, and methylation data to predict cancer survival, outperforming fully supervised baselines [21].

Still, in many cases classical ML models achieve performance comparable to more advanced DL methods [36, 7]. This is partly because complex models involve many

more features, which can amplify noise, whereas simpler models are better understood and easier to interpret.

1.4 Knowledge Gap and Research Questions

While ML models have been applied to cancer survival and drug response, their application to radiotherapy prediction—especially using epigenomic data—is limited.

Moreover, most existing studies face key limitations: small sample sizes, narrow molecular feature sets, unreliable evaluation and limited generalizability beyond in vitro datasets. Few have explored how models trained on multi-omics data from cell lines perform when applied to more clinically relevant in vivo models, such as patient-derived xenografts (PDX).

This dissertation addresses these gaps by focusing on global histone modification profiles as predictive features, exploring their standalone predictive power, their integration with other omics modalities, and their transferability. At the same time, investigating also other dataset (RNA-Seq, Methylation, Proteomics) when they prove more promising. I specifically investigate:

- **Addressing high dimensionality with limited sample sizes** — Which strategies best reduce feature space while retaining predictive signal?
- **Predictive value of global histone modification data** - Does it contain useful information for radiation sensitivity prediction?
- **Multi-modal integration** - Can global histone modification data improve predictions when combined with other modalities? In particular methylation, transcriptomics and proteomics data.
- **Addressing noise in data and labels** - How can I adapt the model, loss and evaluation to account for considerable uncertainty in labels and data?
- **Transferability** - Do models trained on in vitro cancer cell lines translate to more realistic in vivo mouse PDX models?

I address these questions by performing a comprehensive range of experiments on both classical and extended ML models.

1.5 Methods and Experiments

My approach builds on recent advances in multi-omics machine learning while emphasizing robust, reproducible testing to ensure results are transferable to more realistic settings. I prioritize classical machine learning models (e.g., kernel ridge regression, XGBoost, support vector regressors), as these often match or outperform deep learning methods on small-sample, high-dimensional datasets.

To address high dimensionality and improve signal-to-noise ratio, I use multiple complementary strategies:

- *Feature grouping and reduction*: Group genes into biologically informed modules and apply dimensionality reduction (e.g., PCA, low-rank encoders).
- *Imputation of missing values*: Apply a MOSA variational autoencoder to recover missing features and expand usable dataset size.

We then perform modality fusion in several ways:

- *Early fusion*: Concatenate reduced features from histone modifications, methylation, and transcriptomics.
- *Middle fusion*: Use a multi-layer perceptron with *separate modality heads* before merging into shared layers. This also includes Bayesian Neural Network.

Core experiments include:

- *Standalone evaluation* — Quantify predictive performance of global histone modification data using regression models against AUC radiosensitivity scores.
- *Multi-omics performance* — Measure performance gains when histone data is combined with methylation and transcriptomics.
- *Cross-domain testing* — Train on CCLE cell line data and test on PDX model datasets to assess transferability.

To tune the results, and address appearing performance shortcomings, I have also considered:

- *Probabilistic models* that can account for both model and label uncertainty. This also produces uncertainty estimation showing model's confidence in prediction.
- *Classification* (binary/multi-class) that can push models to distinguish between high and lowly sensitive samples.

- *Sample weighting* - Various weighting strategies and their impact on both training and test fit analysed.
- *Alternative evaluation metrics* With many models fitting to the bulk of the labels, we needed to consider alternative metrics including weighted samples and classification.

By combining methodological rigour with multi-omics integration, these experiments aim to establish whether histone modifications are both a *novel predictive modality* and a *complementary source of information* for improving radiosensitivity prediction. At the same time, using other modalities to find best performing models and understand the nature of problem at hand.

Chapter 2

Materials and Methods

I analyze four modalities: global histone modification, RNA-seq, methylation, and proteomics. I employed three dataset variants: CCLE (DepMap data), MOSA-imputed data, and GSVA-derived gene modules. Both classical machine learning models and multi-modal neural networks were applied. To ensure reliable estimates, all training and model selection were performed within a nested cross-validation pipeline. Performance is reported as R^2 against the AUC radiosensitivity response. An overview of this workflow is shown in Figure 2.1.

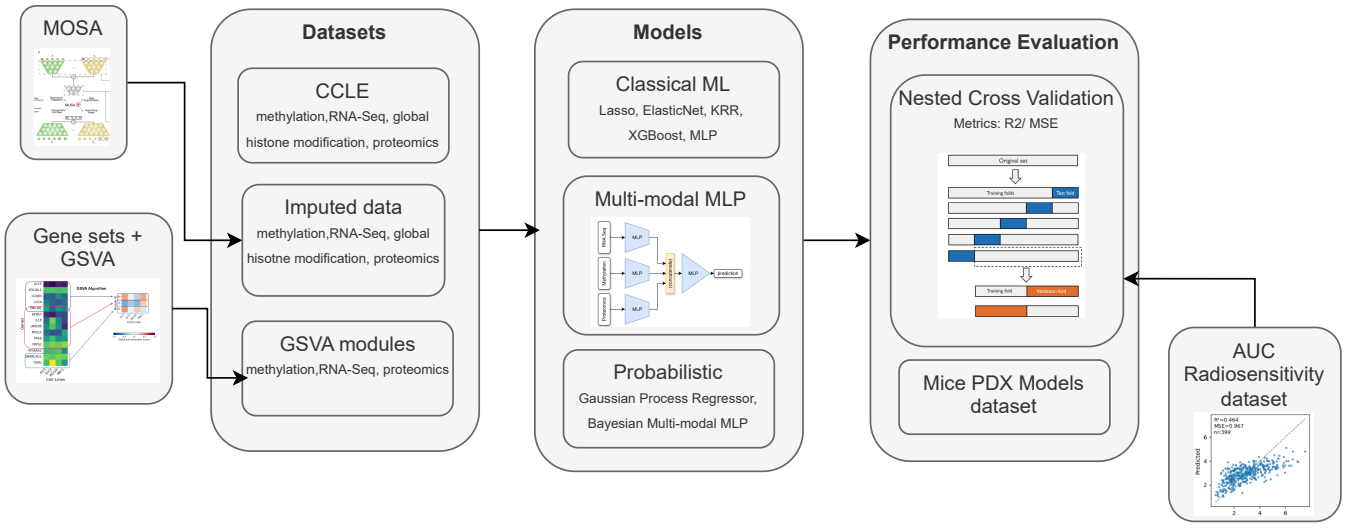


Figure 2.1: Graphical overview of the materials and methods used.

2.1 Datasets

CCLE data I use molecular profiles from the 2019 release of the Cancer Cell Line Encyclopedia (CCLE) [19], including: (i) global chromatin profiling with 42 histone H3 modification features, (ii) RNA-seq gene expression (61,958 features), and (iii) promoter methylation levels (TSS ± 1 kb, 17,181 genes) from RRBS data. The 2019 release was chosen to align with the sampling period of the RadioGx radiation response dataset, minimizing temporal and batch discrepancies. All cell lines are annotated with primary site (eg. lung, bone), histology (eg. neuroblastoma, carcinoma), and subhistology.

Imputed DepMap data I use imputed data generated by the MOSA model [6] (see Section 2.2), which was applied to DepMap datasets to fill in missing values and produce a full set of 1,500 cell lines with complete profiles for all datasets of interest. For RNA-seq, DNA methylation, and proteomics, I directly use the official MOSA imputations [1]. Histone modifications were not part of the original release, so I generated imputations by re-training the model with the CCLE histone data.

Patient-derived xenograft model data To assess the transferability of the findings from cell lines to more clinically relevant systems, I used a panel of glioblastoma patient-derived xenograft (PDX) mouse models [49]. This panel comprises 96 PDXs derived from both primary and recurrent tumours, representing a broad spectrum of glioblastoma cases. The dataset includes RNA-seq, DNA methylation, and gene mutation profiles. The dataset does not provide AUC radiation response, instead the PDX tumors were assigned to treatment groups receiving placebo, radiotherapy (20Gy in 10 fractions), temozolomide, combined radiotherapy and temozolomide, or bevacizumab. In my study, the main response variable is the *Radiotherapy Response Ratio* (RT Ratio), which compares survival in radiotherapy-treated mice to matched controls.

RadioGX Radiation Sensitivity Labels The radiation sensitivity labels (ground truth) are taken from the RadioGx R package [34] (version 2.12.0), which fits a linear–quadratic (LQ) model to dose–response data and then calculates the area under the fitted survival curve (AUC) for each cell line. The LQ model describes the surviving fraction S of cells after a given radiation dose D:

[1] The imputed datasets were obtained from https://doi.org/10.6084/m9.figshare.24562765

$$S(D) = \exp(-\alpha D - \beta D^2),$$

where α and β are constants (with respect to the sample) reflecting the cell's sensitivity to these processes The AUC of this curve provides a single continuous measure of radiosensitivity across multiple dose levels. Higher AUC values correspond to radiation-resistant samples, while lower AUC values indicate sensitivity. In this study, AUC values (ranging from 0.5–7.5) are used as the primary response variable and are not standardized prior to modeling.

2.2 MOSA - Multi-Omic Synthetic Augmentation

Multi-Omic Synthetic Augmentation (MOSA) [6] is an unsupervised deep learning framework designed to integrate and enhance multi-omic datasets from the Cancer Dependency Map (DepMap). It uses a variational autoencoder (VAE) with separate encoders for each omic data type and conditioning on genetic context. This setup enables MOSA to impute missing values and generate synthetic profiles, expanding the number of cell lines with complete data. The original model was trained on seven datasets (omics data and drug response data), and has been shown to improve statistical power in dependency mapping, drug resistance prediction, and cancer subtype clustering.

For my work, I re-trained the MOSA architecture from scratch using the default parameters, except I disabled the KL divergence term in the loss (original weight: 10^{-4}), which otherwise caused numerical instabilities. Histone modification data was incorporated by adapting the `CLinesDatasetDepMap23Q2` class to handle histone features alongside other modalities. I matched cell lines by Broad ID to the Sanger Model ID, generated imputations in the same manner, and then replaced imputed values with original measurements wherever available. This ensured that only genuinely missing values were imputed. The cell lines that were not previously present in histone data, but were entirely imputed, were later excluded.

2.3 Data Preprocessing

Biological datasets often present a challenging combination of high dimensionality and missing values. For example, the methylation dataset contains approximately 17,181 features but only 843 samples, while histone modification has 42 features

and RNA-seq has over 60,000. Many machine learning algorithms require complete data matrices, making the handling of missing values a critical preprocessing step. I considered several approaches, including:

- Dropping samples with missing values.
- Dropping features with missing values.
- Imputing missing values (e.g., filling with zeros or model-based imputation).

In most experiments, I adopt the simplest approach—dropping samples with missing values—unless otherwise specified. The use of MOSA-imputed datasets is particularly advantageous, as it provides complete profiles for all cell lines without the need for further processing.

Matching Across Modalities All datasets were aligned to the RadioGx radiation sensitivity labels by cell line names. In practice, I truncated extended IDs (e.g., RVH421_SKIN → RVH421) and matched them across datasets. Only samples with both molecular features and AUC labels were retained for training and evaluation.

Dataset-specific Cleaning For methylation, I mapped CpG regions to gene loci, averaged values for the same gene, and aggregated duplicate entries. CpG regions are short stretches of DNA where a cytosine (C) is followed by a guanine (G); they are commonly measured because their methylation is linked to whether nearby genes are active. RNA-seq data were filtered to remove duplicate gene entries, retaining the first occurrence. Histone modification data was cleaned by dropping features that had at least 10% of values missing. In the whole dataset (reduced to samples matching response data), 2% of values are missing, with 52% of rows having a missing value and 78% of features having missing value for at least one sample - the preprocessing step however, by dropping only 4 reduces the missing rate to 0.3%, and these appear only in 0.9% of samples.

Scaling For certain experiments, I applied per-feature standardization (zero mean, unit variance) using `StandardScaler` from `scikit-learn` (version 1.5.2). Scaling was always performed within the training folds during cross-validation to avoid data leakage. In many cases, I left values unchanged because some datasets were already preprocessed (e.g., z-scored or log-transformed) or because preserving the original measurement scale was biologically meaningful.

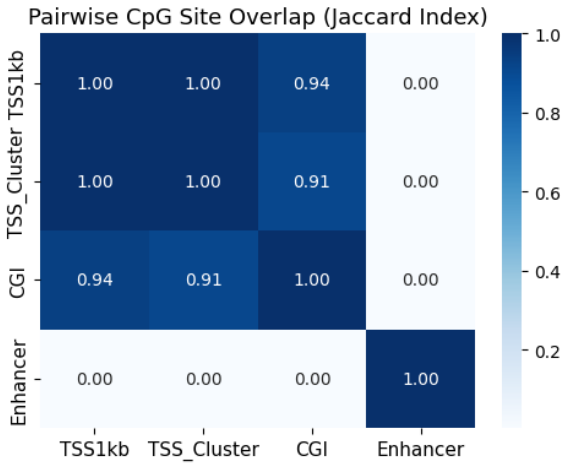


Figure 2.2: Heat map of CpG site overlap across CCLE methylation datasets. Values represent the proportion of CpG sites in one dataset that are also present in another. Enhancer regions contribute the largest set of additional CpGs.

Table 2.1: Properties of the methylation datasets considered. Last column reports percentage of the dataset's genes that are part of the mean methylation (m.m.) dataset.

dataset	**(min, max)**	**mean**	**median**	**genes**	**coverage with m.m.**
mean methylation	(0.0, 0.99)	0.47	0.53	20.248	100%
mice PDX	(0.0, 0.99)	0.36	0.27	25.883	96%
CCLE	(0.0, 1.0)	0.20	0.01	17.100	60%
CCLE ext.	(0.0, 1.0)	0.36	0.33	22.516	74%

2.3.1 Preprocessing of methylation datasets for mean global methylation analysis

To enable replication of the global mean methylation analysis in [12], I prepared methylation datasets from CCLE and PDX models for comparison with the published data. Different datasets report distinct sets of DNA methylation regions:

- The CCLE data (and the MOSA-imputed) I use is restricted to the *promoter methylation*. [2]
- The Mean Global Methylation work uses *genome-wide DNA methylation* using the Illumina 450k array or Illumina EPIC (850k) array.

[2]Data is available at the NCBI portal with accession code GSE240704 https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE240704

- PDX model data uses the *genome-wide DNA methylation* using Illumina EPIC array (850k).

Extracting Genes Used in the Original Publication The mean methylation paper provides the normalized beta matrix—that is, values are in range [0,1] with rows being CpG sites and columns being samples. I use the `BiocManager` (version 3.21) and `minfi` R package and load the Illumina EPIC array annotation package for the hg19 genome build. This provides the probe-to-gene mapping and genomic locations. In this way, I can extract which genes have been covered and determine the total number of covered genes, specifically promoter genes. This list of genes can then be used to replicate results for the Mice PDX data and CCLE data where the data is represented on a per-gene basis.

Extending CCLE Methylation Beyond Promoter Region The publicly available CCLE methylation data are typically limited to CpG sites within promoter regions (1 kb upstream of the TSS, TSSS1kb). To obtain a broader view, I include additional CCLE RRBS datasets covering promoter CpG clusters, CpG islands, and VISTA enhancers. These datasets are not provided at the gene level, so I first process the region-level data: for loci directly annotated with gene names, the mapping is used as is; for the remaining regions, genomic coordinates are extracted and overlapped with hg19 gene annotations. For each gene, methylation values are then aggregated across mapped regions using a weighted mean based on coverage. This procedure yields an extended, gene-level methylation matrix that integrates promoter, island, and enhancer information. The heat map in Figure 2.2 summarizes the degree of CpG site overlap across the four CCLE methylation datasets (TSS1kb, promoter clusters, CpG islands, and enhancers). Values represent the proportion of CpG sites in one dataset that are also present in another.

Overlap Between Datasets By inspecting the mean, median, and range of beta values, I confirm that all datasets report raw methylation levels on the same scale as the original study, without additional normalization (Table 2.1). The extended CCLE dataset in particular shows distributions that are more similar to those used in the mean methylation publication.

In terms of gene coverage, the mean methylation dataset contains 20,248 genes, of which 19,580 are also present in the mouse PDX data. The promoter-level CCLE data has a more limited overlap, with 12,300 genes (out of 17,100), whereas the extended

CCLE dataset captures a broader set of 15,000 genes (out of 22,516).

Together, these preprocessing steps yield consistent gene-level methylation matrices across datasets, which were then used in downstream analyses (see Section 3.1).

2.4 Training and Validation Pipeline

I have placed strong emphasis on establishing a reliable and comparable testing procedure, drawing on the configuration and data processing setup provided with P-NET [13], which I adapted and extended for our own validation framework.

2.4.1 Data Splitting and Cross-Validation Strategy

To ensure robust model evaluation and avoid overfitting, I adopted a **nested cross-validation** framework. This approach consists of two levels of K-fold cross-validation: an outer loop for estimating generalization performance, and an inner loop for hyperparameter tuning.

Outer Cross-Validation

The full dataset was split into $K = 5$ outer folds. In each iteration of the outer loop, one fold was held out as a *test set*, while the remaining four folds were used for training and validation. This procedure was repeated five times such that each sample appeared exactly once in the test set. Model performance was evaluated independently on each outer test fold, and the results were later aggregated (mean and standard deviation) to report final metrics.

Inner Cross-Validation and Hyperparameter Tuning

For each outer training set, I conducted a $K = 5$ **inner cross-validation** to tune model hyperparameters. The training data were split into five inner folds: in each iteration, four folds were used for training, and one fold was used for validation. A grid search was applied over a range of hyperparameter configurations, and the configuration yielding the highest average validation performance (measured using the coefficient of determination, R^2) across the inner folds was selected.

Validation and Test Strategy

The inner validation folds were used exclusively for model selection, while the outer test folds were held back entirely for final performance assessment. No portion of the test data was used during training or validation. Once the best hyperparameters were determined via the inner loop, a final model was trained on the full outer training set and evaluated on the corresponding outer test set. I did not use an additional validation subset within the test fold itself.

This nested setup guarantees an unbiased estimate of model generalization, while also enabling fair model comparison across different algorithms and feature selection strategies.

2.4.2 Performance Metrics

To assess model performance, I reported two standard regression metrics: the coefficient of determination (R^2) and mean squared error (MSE). The metrics are defined as follows. Given true target values y_i and corresponding predictions $\hat{y}_i$, for $i = 1, \ldots, n$:

Mean Squared Error (MSE) :

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Coefficient of Determination (R^2) :

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ is the mean of the observed targets.

The R^2 score represents the proportion of variance in the target variable that is predictable from the input features. A value of 1 indicates perfect prediction, while values closer to 0 or negative suggest poor model performance. Due to its scale-invariance and direct interpretability in terms of variance explained, I used R^2 as the primary metric for model selection and comparison.

To better evaluate performance in more complicated settings, I consider some additional performance evaluation functions. The ϵ-intensive loss attempts to take into account the label noise - similarly to Negative Loss Likelihood. While Sperman's ρ and AUROC enable performance evaluation when the available response data is on a different scale than the responses.

ϵ-intensive loss (Support Vector Regressor loss)

$$L(y, \hat{y}) = \max(0, |y - \hat{y}| - \epsilon)$$

If the prediction $\hat{y}$ is within the ϵ-tube of the true value y, the loss is zero. If the prediction is further than ϵ from the true value, the loss grows linearly with distance.

Sperman's ρ (rank correlation) It measures correlation between the *ranks* of true and predicted values. It enables testing whether predictions preserve the ordering of samples - it is scale independent. Values fall in range $[-1, 1]$ with 0 corresponding to no rank correlation and 1 being a perfect relation.

AUROC (sensitive vs. resistant, split by median) Measures ability of predictions to discriminate between two groups (here: sensitive vs. resistant, defined by splitting true values at the median). Intuitively, it is the probability that a randomly chosen "sensitive" sample scores lower (more sensitive) than a "resistant" one. The values fall in the range $[0, 1]$ with 1.0 being a perfect separation and 0.5 being random. This can act as conversion of output into a binary classification problem.

2.5 Dimensionality Reduction Strategies

Dimensionality reduction is an important tool to address the high dimensionality of molecular datasets, particularly when the number of features greatly exceeds the number of samples. In this work, I explored three general approaches:

- Classical dimensionality reduction — such as Principal Component Analysis (PCA) and Truncated Singular Value Decomposition (Truncated SVD).
- Feature selection — using variance thresholding and statistical tests.
- Grouping features into modules - aggregating related features based on prior biological knowledge or statistical similarity.

This step was particularly important when combining histone modification data with other modalities. Since histone data contains only 42 features, concatenating it directly with a modality containing thousands of features (e.g., methylation or RNA-seq) could cause its contribution to be overwhelmed. To address this, I applied dimensionality reduction to the higher-dimensional modality before concatenation.

All dimensionality reduction was performed *within each training fold* during cross-validation to avoid data leakage.

Classical Dimensionality Reduction PCA and Truncated SVD are widely used linear methods for projecting high-dimensional data into a lower-dimensional space while preserving as much variance as possible. I have used the `sklearn` implementations using 50 components.

Feature Selection Feature selection reduces dimensionality by retaining a subset of the most informative features, which can help interpretability, and remove noise or redundant variables.

- I used `SelectKBest` with `score_func` = `f_regression` and $k = 200$.
- I also experimented with removing features below a variance threshold of 0.1.

Feature selection was performed independently for each training fold to avoid information leakage from the test data.

2.5.1 Modules and Gene Set Aggregation

One way to incorporate prior knowledge into machine learning models is to group genes into modules—sets of related genes that are likely to participate in the same biological process. In genomics, this means taking thousands of molecular measurements and summarizing them into a smaller number of biologically meaningful categories. These modules can be based on curated pathway databases (as in P-NET [13]) or derived directly from large-scale experimental data. For example, DeepProfile identified gene modules from leukemia datasets and used them to predict drug response [10].

In this thesis, I use the genome-wide atlas of co-essential modules from [50], built from 485 CRISPR knockout screens in the DepMap Achilles dataset. The authors used a statistical method that accounts for similarities between cell lines, reducing false matches compared to simpler correlation-based approaches.

I experimented with using all identified groups of genes (5.229 modules), a filtered set of highly enriched ones ($\geq$ 100-fold enrichment with Bonferroni-corrected $P < 0.05$, 1.269 modules), or take the top 500 most-enriched. With these modules, I computed GSVA activity scores that summarize activity levels within each set

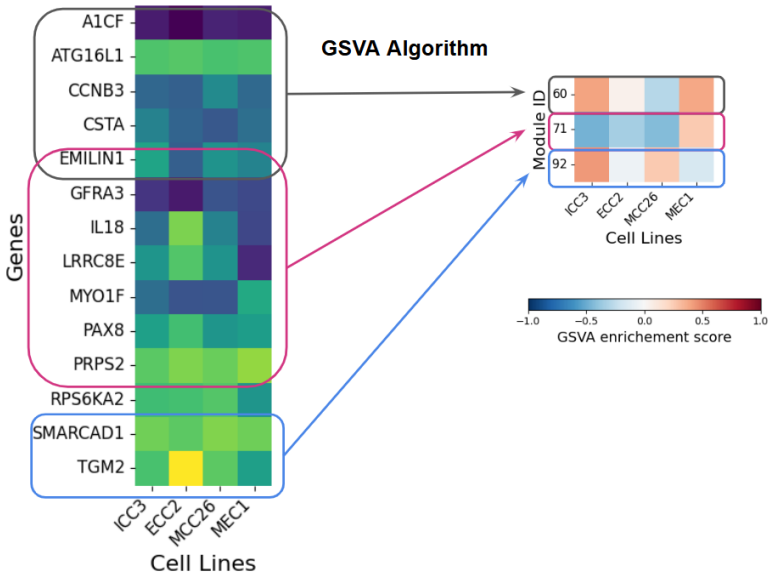


Figure 2.3: **GSVA workflow.** Genes are grouped into modules and aggregated with GSVA to yield per-module enrichment scores across cell lines.

(Fig. 2.3). This produces compact features that reflect coordinated functional activity rather than individual gene measurements. Alternative to GSVA scoring metrics have been considered, but proved less resilient (see Section 3.3).

GSVA (Gene Set Variation Analysis) is a method used in biology to turn a group of genes (such as a pathway or a module) into a single score that reflects how active that group is in a sample [24]. It works by ranking genes within each sample and applying a non-parametric transformation to compute enrichment scores for predefined gene sets. Unlike traditional gene set enrichment methods that are typically applied at the end of an analysis, GSVA allows for pathway-centric modeling at an early stage. In my analysis, I used the established R implementation of GSVA (version 2.2.0), available through the Bioconductor project [18] (version 3.21).

2.6 Combining multiple modalities

A straightforward approach to combining multiple data modalities is to simply *concatenate* their feature matrices. While convenient, this method has notable drawbacks: features may be on different value scales (which can affect scale-sensitive models such as SVMs or regularized linear models), dimensionality increases substantially, and merging a lower-dimensional dataset with a much larger one can lead to imbal-

ances. In gene-based datasets such as methylation and RNA-seq, naive concatenation can also obscure the correspondence between features, losing biologically meaningful relationships. Some approaches, such as biologically informed neural networks, explicitly encode these relationships to preserve interpretability.

An alternative is *mid-fusion*, where the architecture includes separate modality-specific "heads" that extract representations for each dataset before merging them in an intermediate layer. The combined representation is then used by the rest of the model, allowing gradients to propagate through all modality-specific branches. This approach is particularly useful when working with datasets of varying size or feature structure. An example of such fusion would be the MOSA model mentioned earlier.

A *late-fusion* strategy instead trains separate models on each modality and then combines their predictions, for example by taking the mean, median, or a weighted average. This method can be advantageous when modalities are heterogeneous or when sharing representations across them is not straightforward. Moreover, if multiple weak regressors achieve similar performance, late fusion can have a "boosting-like" effect, where aggregating their outputs leads to more robust and accurate predictions.

2.7 Baseline Models

I consider a range of established models ranging from widely used classical machine learning approaches, through gaussian models and finally, the P-Net architecture.

2.7.1 Classical Machine Learning methods

I explored a range of established machine learning methods, using open-source library implementations for robustness. Unless otherwise specified, the models were from `scikit-learn` library (version 1.5.2).

LASSO Regression A linear regression model with an L1 penalty, which encourages sparsity in the coefficient vector by driving weights to zero. It is particularly effective when only a subset of features is expected to be relevant. I tuned the regularization strength over penalty term $\alpha \in \{0.01, 0.05, 0.1, 0.3, 0.5, 1.0\}$.

Elastic Net This model combines the strengths of both the L1 (sparsity) and L2 (multicollinearity handling) regularization. It uses a mixing parameter β that determines how much the model behaves like LASSO ($\beta = 1$) and how much like

ridge regression ($\beta = 0$). It has been shown to be particularly useful in problems when number of features is larger than the number of data points. I searched over the penalty term $\alpha \in \{0.001, 0.05, 0.1, 0.3, 0.5, 1.0\}$ and the mixing parameter $\beta \in \{0.0, 0.05, 0.1, 0.2, 0.5, 0.7, 0.9, 1.0\}$.

Kernel Ridge Regression Combines the ridge regression method with kernel trick using a selection of kernels (eg. sigmoid, Radial Basis Function (RBF), polynomial). Recall that ridge regression is a linear model with L2 penalty (least squares solution with L2 regularization). KRR has a regularization parameter α and kernel-specific parameters. One of limitations in practice of using KRR is that it can be expensive to compute the kernel matrix on large datasets - then randomization methods such as Nyström method need to be used [17]. However, this is not the case for datasets used in this study - having ~ 500 data points doesn't raise the issue, enabling us to efficiently use this strong method. Important considerations however are its sensitivity to kernel selection its parameter and regularization parameter choices - hence a wide search is particularly important. I tuned the parameters over:

- polynomial kernel - degree $\in \{1, 2, 3\}$, $\alpha \in \{0.01, 0.1, 0.5, 1, 3, 5, 10, 50\}$
- RBF and sigmoid kernels - $\gamma \in \{0.001, 0.005, 0.01, 0.03, 0.05\}$, $\alpha \in \{0.1, 1, 10\}$

XGBoost A gradient boosting framework that builds an ensemble of decision trees sequentially, where each new tree corrects the residual errors of the previous ones. Regularization terms in its objective function help control overfitting, making it well-suited for high-dimensional, heterogeneous datasets. I searched max depth $\{3, 5\}$, learning rate $\{0.005, 0.01\}$, number of estimators $\{300, 500, 800\}$ and subsample and column sample both 0.8.

Support Vector Regressor A regression variant of support vector machines that fits a function within a margin of tolerance ϵ, while penalizing deviations linearly. Using kernel functions (e.g., RBF, polynomial) allows it to model complex non-linear relationships while maintaining sparse support in the dual representation. I tested the polynomial kernel with degree $\in \{1, 2, 3\}$, γ set to 'scale' and $C \in \{1000, 100, 10, 2, 1, 0.2, 0.1, 0.02, 0.01\}$.

2-Layer MLP Using the `MLPRegressor` model, I did an architecture search over first layer $\{64, 32, 16\}$ and second layer dimensions $\{32, 16, 8\}$. I am using a very

strong weight decay, parameter $\alpha \in \{0.1, 1.0, 5.0\}$ (note, the default value is normally 0.001) as it helps push many of the weights to zero, which is required with the small amount of data but high input dimensionality. Using learning rate scheduler (`n_iter_no_change` set to 5) enables me to set a high number of epochs 300 with these parameters performing well across varied datasets. I have decided not to use early stopping as it is based on randomly set aside part of the dataset - however with such small dataset this is often unrepresentative and leads to errors.

2.7.2 Classical Probabilistic models

Gaussian Process Regressor This model employs a Gaussian process as a prior over functions and updates this prior with observed data during training to form a posterior distribution. An important property of Gaussian process regression is its ability to quantify predictive uncertainty. Furthermore, through the choice of kernel functions, the model can flexibly adapt to data exhibiting different degrees of smoothness and complexity.

Using the `GaussianProcessRegressor` from the `sklearn` library, I explored a variety of kernels, including:

- WhiteKernel — models i.i.d. Gaussian noise; typically combined with other kernels.
- RBF — smooth, stationary kernel that introduces nonlinearity.
- RationalQuadratic — a scale mixture of RBFs; good when the function varies over multiple length scales.

I also considered their combinations (e.g., through addition) to capture more complex structures in the data. Using `normalize_y` set to `True` and number of optimizer restarts set to 2. The alpha parameter (noise estimation that can be passed explicitly, or inferred by the model) was experimented with in a range of $0.1 - 02$.

2.7.3 P-NET - biology-informed neural network

P-NET is a biologically-informed neural network designed to predict clinical outcomes, such as metastasis in prostate cancer, using genomic data [13]. Unlike traditional dense networks, P-NET uses a sparse, feedforward architecture where each node represents a biological entity (e.g., a gene or pathway) and each edge represents a known biological relationship, based on curated databases. This structured

design leads to a dramatic reduction in model complexity and number of parameters, improving interpretability and computational efficiency. Importantly, the network architecture is flexible and customizable, built automatically from user-provided pathway definitions (e.g., in .gmt format), allowing integration of domain-specific biological knowledge. I used it as a baseline on the methylation dataset with the default training parameters.

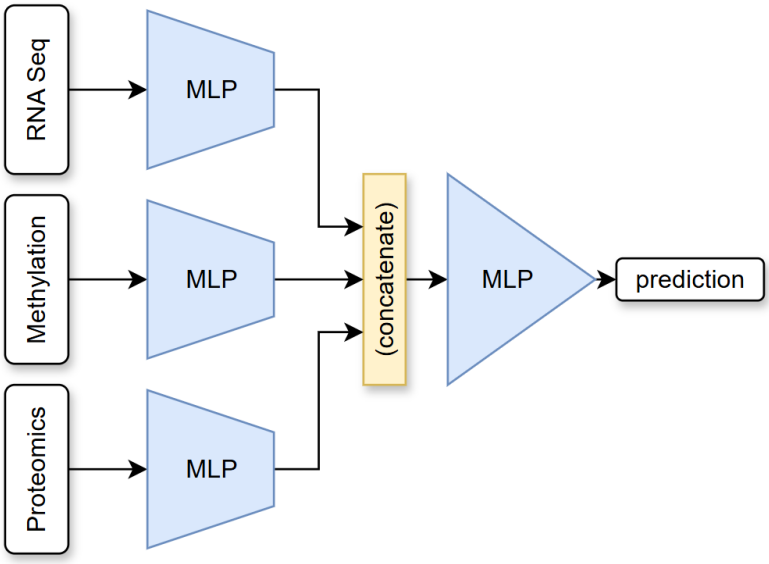


Figure 2.4: Mid-Fusion approach using modality-specific MLP heads, concatenation of latent representations and a fusing MLP

2.8 Mid-Fusion using Neural Networks

I implemented a mid-fusion architecture in which each data modality is processed by its own modality-specific head—a multi-layer perceptron (MLP) that learns a low-dimensional feature representation. The outputs of these heads are concatenated and passed to another MLP for final regression. Gradients flow through the entire model, allowing joint optimization of all parameters. An architecture diagram can be seen in Figure 2.4.

The model is implemented from scratch in PyTorch using `nn.Linear` layers, ReLU activations, and dropout for regularization. Training uses mean squared error loss (`MSELoss`) and the Adam optimizer. Architecture parameter search goes over the depth of modality-specific heads (2-3 layers) and hidden size (16, 32 or 50 units), while the combining MLP has two layers with search done over the ([32, 8], [64,

16] and [128, 32]) combinations. This results in an architecture small enough not to overfit but still representative enough to produce meaningful results. I have also found that incorporating a learning rate scheduler and strong weight decay have improved performance - regularization is particularly important in a low-data, high-dimensionality setting especially when there are as many parameters as in an MLP. In the plateou scheduler setting, I use `ReduceLROnPlateou` from the `optim` library with scheduler factor 0.5, patience 2 and early stopping patience 5. Weight decay has proven to be most effective when kept high - in the $0.001 - 0.2$ range.

2.8.1 Bayesian-style architecture

To incorporate model uncertainty, I build a Bayesian-style NN, implemented in PyTorch (version 2.7.1). It can work both with a single modality, as well as incorporate multiple ones - analogously as described in above mid-fusion MLP. Gal and Ghahramani [14] showed that applying dropout at inference time can be interpreted as approximate Bayesian inference in deep Gaussian processes. This approach, known as Monte Carlo (MC) dropout, enables us to efficiently estimate epistemic (model) uncertainty without modifying the network architecture or training procedure.

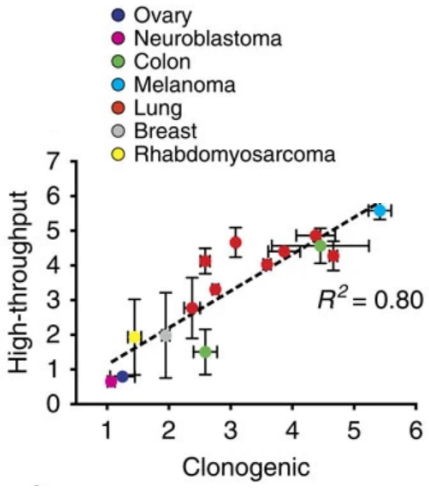
In addition to epistemic uncertainty, the regression setting also includes aleatoric uncertainty arising from measurement noise in the training labels. I account for this in the loss by using a negative log-likelihood (NLL) with known label noise σ (e.g. the reported standard error of the mean, SEM). Specifically, for N samples $\{(y_i, \hat{y}_i)\}$ the loss is

$$\mathcal{L}_{\text{NLL}}(y, \hat{y}; \sigma) = \frac{1}{N} \sum_{i=1}^{N} \left[\frac{(y_i - \hat{y}_i)^2}{2\sigma_i^2} + \tfrac{1}{2} \log\bigl(2\pi\sigma_i^2\bigr) \right], \tag{2.1}$$

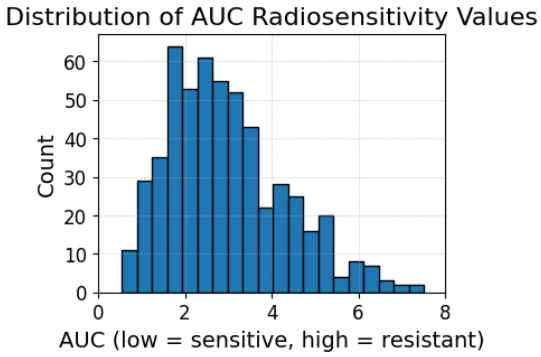
where y_i is the true label, $\hat{y}_i$ is the network prediction, and σ_i is the known label noise. In most cases, I assume a fixed $\sigma_i = \sigma$ across samples. This formulation allows us to disentangle epistemic uncertainty (via MC dropout) from aleatoric uncertainty (via label noise in the NLL loss).

For the **Bayesian MLP** I used dropout probability of 0.1, 20 epochs, fixed noise estimation $\sigma = 0.1$ and different possible sets of layers $(32, 4), (32, 16, 4), (128, 32)$.

For the **Mid-fusion Bayesian MLP** dropout probability is 0.1, number of MC samples (to generate uncertainty prediction) 30, default noise estimation $\sigma = 0.1$ and 20 epochs. I used as modality-specific heads 3 layers of dimension 32 and the combining MLP with two layers $(128, 32)$.



(a) **Estimating AUC uncertainty from published error bars.** High-throughput AUC estimation (2 samples per each cell line) with mean±s.e.m. error bars, adapted from Yard et.al. 2016.



(b) **Distribution of AUC values in the RadioGX dataset.** Most samples cluster in the 1.8–3.2 range, producing a skewed distribution.

Figure 2.5: Characterizing label noise and imbalance in AUC values. (a) Estimation of AUC uncertainty based on published error bars. (b) Empirical distribution of AUC values used in this study.

2.9 Addressing factors limiting performance: Imbalanced distribution and label uncertainty

Accurately predicting radiation sensitivity is restricted by both noise in the response labels and imbalance in their distribution. First, label noise can be quantified directly from the experimental error bars reported in the source paper of RadioGX AUC responses [53]. By inspecting the high-throughput assay with n=2 replicates (Fig. 2.5a), I estimate the s.e.m. across cell lines as mean $\approx$ 0.35, median 0.20, range 0–1.25. Because error bars are reported as s.e.m., converting to standard deviation (SD = s.e.m.$\times\sqrt{2}$ for n=2) gives mean $\approx$ 0.48, median 0.30, and values up to 1.70. This highlights the substantial uncertainty in reported AUC values.

Second, cell lines themselves are unstable. They accumulate genetic and transcriptional changes over time, which can alter drug responses and reduce reproducibility. Ben-David et al. [4] demonstrated extensive diversity across widely used lines, showing that even strains of the same cell line can diverge in gene expression and drug sensitivity.

Finally, the label distribution itself is highly skewed, leading to an imbalanced

labels problem. As shown in Fig. 2.5b, most samples fall within the 1.8–3.2 range, with extreme sensitivities underrepresented. The key objective is to distinguish between low- and high-sensitivity samples, but this is complicated by the combination of noise, class imbalance, and a weak underlying signal. In the following, I explore strategies to mitigate these challenges.

2.9.1 Classification as a way to handle label noise

Branson et al. [5] propose binarizing treatment response values under the hypothesis that this reduces experimental noise. Their models (BinaryET and BinaryCB) show improved performance when predicting binary drug response, suggesting that classification can align more closely with learnable biochemical signals. Indeed, many studies adopt classification strategies to boost results.

However, this approach was less effective for us due to weak separation in the response values. Still, it allows us to evaluate how well the model distinguishes between regions of the response distribution.

I experiment with using both binary classification (AUC labels split at the median value 2.7) and a four-class setup defined by AUC intervals [0,2), [2,4), [4,6), and [6,8). Overall and per-class accuracy is reported using `accuracy_score` from `sklearn`; overall accuracy in the inner folds is used for model selection. Models include Logistic Regression (LR), a Support Vector Classifier (SVC) with polynomial and RBF kernels, and XGBoost. The SVC grid matches the SVM search described earlier. XGBoost follows the regression grid with the evaluation metric set to `logloss`. For LR, I grid-search $C \in \{0.01, 0.1, 1, 10\}$ with an ℓ_1 penalty, the `saga` solver, and up to 5000 iterations.

In the multi-class setting I also use class-balanced weighting. For `sklearn` models I set `class_weight='balanced'` (or equivalently pass `sample_weight` to XGBoost, where I compute the weights). This assigns a weight w_c to each class so that $w_c N_c$ is constant across classes $c \in \{1, 2, 3, 4\}$; explicitly,

$$w_c = \frac{N}{|C|\, N_c},$$

where N is the total number of training samples and $|C| = 4$ is the number of classes. Thus each class contributes equally to the loss.

2.9.2 Sample weighting strategies

Most `sklearn` models allow incorporating *sample weights* via the `sample_weight` argument in the fitting function. The purpose of weighting here is to emphasize the extremes (highly and low-sensitive samples), rather than letting the model fit mainly to the bulk of labels. I evaluate two weighting strategies:

Percentile-distance weighting. Weights are based on the distance of each label's percentile from the median (50th):

$$w_i = (|F(y_i) - 0.5| + \epsilon)^\alpha,$$

where $F(y_i)$ is the empirical CDF, α controls tail emphasis, and ϵ prevents zero weight at the median. I explore $\alpha \in \{1.5, 2.0, 3.0\}$.

MAD-scaled power weights. This variant uses a robust, scale-aware metric:

$$w_i = (z_i + \epsilon)^\alpha, \quad z_i = \frac{y_i - \text{median}(y)}{1.4826 \cdot \text{MAD}(y)},$$

where MAD is the median absolute deviation. I explore $\alpha \in \{1.5, 2.0, 3.0\}$.

At each fold, weights are normalized to have mean 1 to maintain loss scale stability.

2.10 Transferability of models to PDX Mice Data

This is the procedure I followed to evaluate models on the PDX data:

1. Preprocess the Mice PDX data. Analyze the scaling used in the original data and repeat the process for the PDX data. If raw values are used in both datasets, keep all the values as is. The scaling is analyzed using minimal, maximal and mean dataset values as well as the details in source publications of the datasets.

2. Fill in the missing values and features in PDX data. Identify the features used in the model training and identify ones missing in the mice data. Fill in the missing features with a mean value from the original data - such imputation encourages neutrality of such missing gene's contribution to the prediction.

3. If the original dataset employs pathway-based modules (e.g., GSVA scores), apply them to the PDX data. Direct recomputation is not straightforward,

since GSVA is dataset-dependent and produces non-comparable values across datasets. To address this, I use a heuristic: replace randomly selected rows of the original dataset with the PDX samples, compute GSVA as usual, and extract the PDX-specific scores. Since the PDX cohort is small (~ 66) relative to the original dataset ($\sim$800 samples), this substitution is not expected to substantially distort the overall data distribution.

4. For each dataset–model combination (e.g., methylation with kernel ridge regression), select hyperparameters based on prior cross-validation results. Rather than tuning separately for each fold, adopt the configuration with the best average performance across folds.

5. Train the model on the whole available original dataset using chosen parameters. Save the trained model.

6. Run the model on the PDX data. First align the features, load the model and note predictions on the PDX data.

7. Evaluate the results by inspecting a scatter plot (predicted AUC vs. true Radiation Sensitivity Ratio) and reference evaluation metrics (R2, MSE). Note though that the radiosensitivity response data for the Mice dataset is the RT Ratio (of survival time) while the predictions are the AUC radiosensitivity value.

Preprocessing of the PDX methylation and RNA-seq data The methylation data has reported the beta values in both the imputed CCLE dataset and mice PDX methylation data, so no processing was required. In the RNA-seq case I had access to only the RPKM counts of the PDX Models RNA data, that is values are in the range [0.0, 5933.0]. While the imputed data has been processed in a different way - RNA-seq dataset used as input to the MOSA model (source: [16]) was TMM normalized and voom-transformed, resulting in data falling in the range $[-12.3, 18.1]$ and mean 3.05. To match the scale and distribution without having access to the raw counts of either dataset, I use the following heuristic: log-transforming the PDX RPKM, then per-gene aligning by quantiles (`normalize.quantiles.use.target`) of the RPKM matrix to the reference MOSA matrix. The resulted processed data has values in range $[-7, 13.4]$ and mean 3.05, which fits the distribution of the original data well enough.

Chapter 3

Results

First, I evaluated how the standalone modalities (histone, methylation) perform on classical ML models. I addressed the missing value problem by running the MOSA encoder-decoder model. Then, I considered dimensionality reduction of gene-based modalities (RNA-Seq, methylation, proteomics), and evalute multi-modal performance. Lastly, I address the imbalanced labels problem and test transferability of results to the PDX models.

3.1 Global histone marks are weak standalone predictors; global *mean* methylation might not be robust

Aim & approach. To test whether global histone modification and methylation predict radiosensitivity on their own, I trained classical ML models under consistent preprocessing. To validate whether global mean methylation indeed predicts radiosensitivity, I reproduced the mean-methylation analyses done by Eckhardt et. al. [12] in both CCLE data and PDX models.

The histone modification dataset contains 42 features and 897 samples, of which 487 include radiation response data. After removing samples with missing values, only 245 remain. Although this reduced subset performed reasonably well (Figure 3.1, (42 features)), the reliability is limited by its small size. The apparent improvement is likely driven by an easier-to-learn subset of samples, suggesting a case of samples *missing not at random*, where missingness reflects underlying sample properties. To address this, I use a reduced 39-feature version of the dataset, in which particularly sparse features were removed, yielding 440 complete samples. This dataset will be

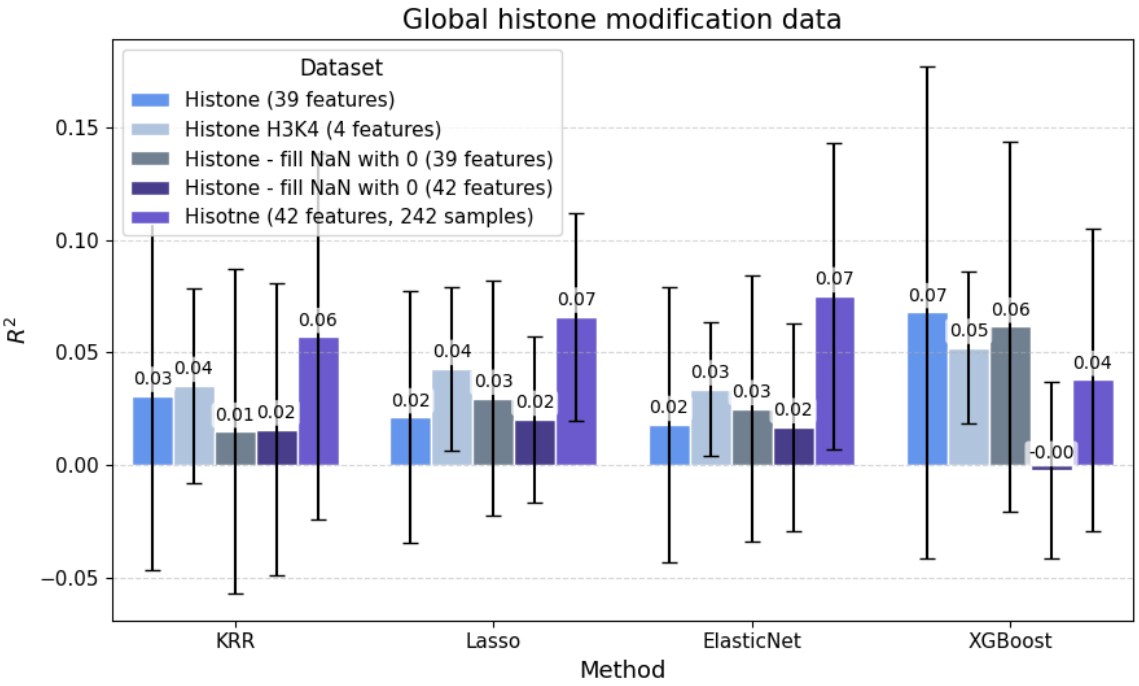


Figure 3.1: Performance of the global histone modifications dataset under different preprocessing strategies. Dropping samples with missing values yields better performance than zero-imputation. The mean and variance of the R^2 metric across folds are reported.

referred to as *Histone* in the following. An alternative strategy was to impute missing values with zeros, which restores the full 487 samples but consistently lowers performance (Figure 3.1, fill NaN with 0). I also explored dimensionality reduction and feature selection methods. Both TruncatedSVD and SelectKBest produced negative R^2 values across all models. However, inspired by prior work [3, 42], I focused on the H3K4 histones (4 features: `H3K4me0`, `H3K4me1`, `H3K4me2`, `H3K4ac1`), which led to an improvement in results. Other subsets, derived both computationally and biologically, were also tested, but did not yield improvements.

Overall results, summarized in Figure 3.1, show substantial variability across runs. Feature reduction helps stabilize variance, but the best-performing model (XGBoost) reaches only about $R^2 = 0.07$, which remains unsatisfactory.

Next, I evaluate the performance of classical ML methods on the CCLE methylation data. Given its high dimensionality (17,124 features), I also consider dimensionality reduction strategies. First, I restrict to the subset of non-protein-coding RNAs (13,294 features) [1]. I then apply:

[1]Non-protein-coding RNAs list: https://www.genenames.org/data/genegroup/#!/group/475

Table 3.1: Model performance on **DNA methylation** data under different dimensionality reduction strategies. Values are R^2 mean ± s.d. across folds.

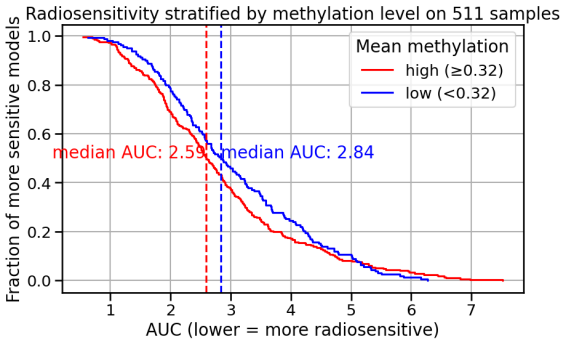
	Original genes	**Trunc SVD 50 comps**	**Trunc SVD 100 comps**	**Feature Selection**	**Variance Thr. 0.1**
Num. Features	13,294	50	100	200	1300
KRR	0.105 ± 0.056	0.153 ± 0.122	$\mathbf{0.168} \pm 0.102$	0.06 ± 0.125	0.132 ± 0.058
ElasticNet	$\mathbf{0.162} \pm 0.082$	0.140 ± 0.109	0.162 ± 0.095	—	0.136 ± 0.038
XGBoost	0.100 ± 0.076	0.129 ± 0.085	0.122 ± 0.076	—	0.100 ± 0.076
Lasso	0.058 ± 0.065	0.148 ± 0.100	0.161 ± 0.099	0.058 ± 0.07	0.052 ± 0.056
P-Net	0.054 ± 0.023				
MLP (5 layers)	0.129 ± 0.059				
MLP (2 layers)	$\mathbf{0.152} \pm 0.096$				

- *Dimensionality reduction:* Truncated SVD with 50 and 100 components
- *Subset selection:* SelectKBest ($k = 200$) and VarianceThreshold ($\theta = 0.1$)

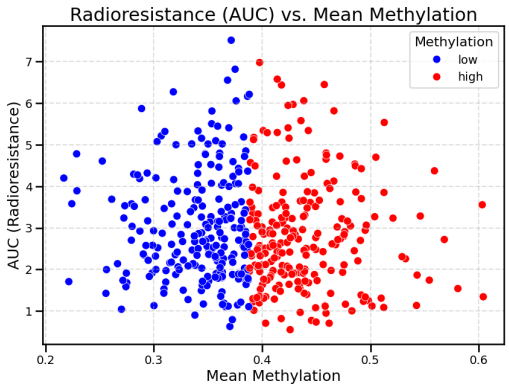
Results are shown in Table 3.1. Truncated SVD improves performance, reaching $R^2 = 0.168$, but with noticeably higher variance. This effect arises because dimensionality reduction is performed within nested cross-validation: depending on the subset of data, the reduced representation can be more or less informative. If reduction were applied outside of cross-validation, performance would appear more stable, but this would risk information leakage. Feature selection performed similarly poorly, with both low R^2 and high variance. Variance thresholding produced the most stable results, though at the cost of reduced performance. As a baseline, I also follow the P-Net paper [13], evaluating P-Net and their multi-layer dense networks on the methylation data. P-Net itself performs poorly, while the best neural model is a 2-layer MLP, achieving $R^2 = 0.152$.

Finally, prior studies suggest that mean methylation levels may predict radiosensitivity in certain cancers [12]. I attempted to reproduce this finding using both the CCLE and the more clinically relevant PDX datasets. It is also known that methylation of the MGMT gene is associated with response to the chemotherapy temozolomide [11, 12]. I replicated this result on the PDX models (as they have TMZ response data available).

I used RadioGX AUC radiation–response data to examine how global mean methylation relates to radiosensitivity. Missing values were set to `NaN` and excluded when computing means. The median mean methylation across samples is 0.35, and the Pseudo-KM plot (Fig. 3.2a) reveals a slight correlation of low methylation and resis-

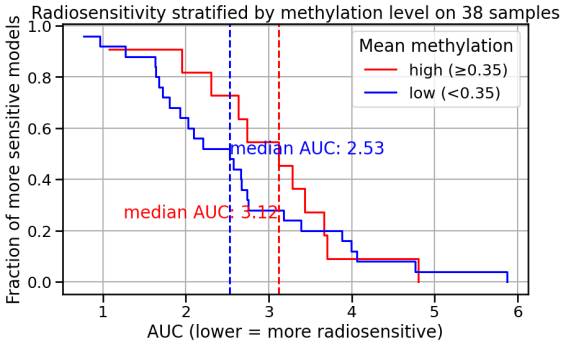


(a) Pseudo–Kaplan–Meier plot shows slight correlation of high methylation and sensitivity.

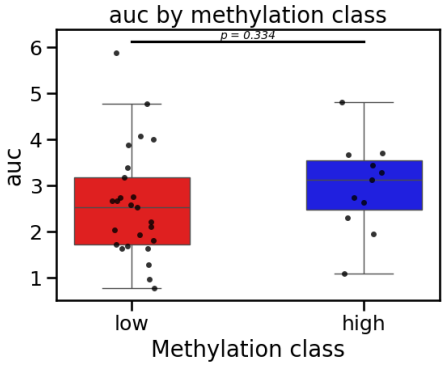


(b) Scatter plot mean methylation vs. AUC. Low and high methylation (by median 0.35).

Figure 3.2: Global mean methylation and radioresistance in CCLE data. (a) Survival-like view of the AUC distribution. (b) Scatterplot of high and low methylation vs. AUC.

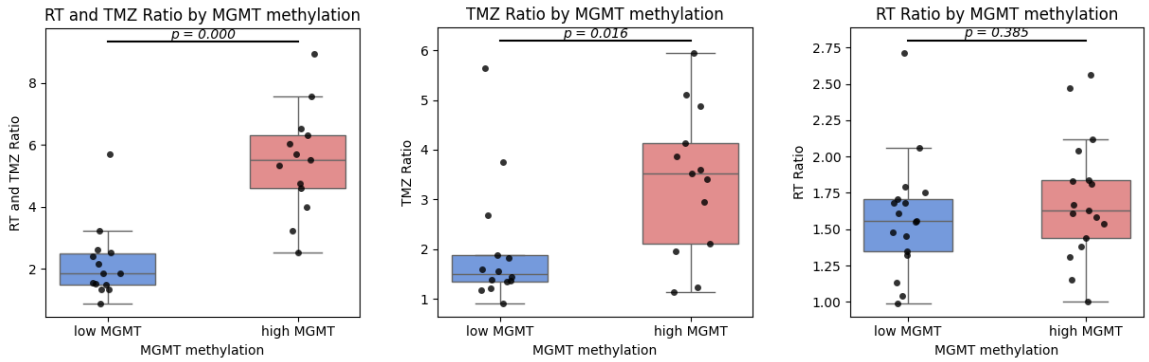


(a) Pseudo Kaplan-Meier plot



(b) AUC by methylation class (split on 0.35); Welch's t-test p shown.

Figure 3.3: Global mean methylation and AUC response for ovary cancer. The result is not significant.



(a) RT+TMZ (b) TMZ alone (c) RT alone

Figure 3.4: Correlation between *MGMT* promoter methylation and TMZ therapy response in PDX models. (a) Combined RT+TMZ treatment. (b) TMZ treatment alone. (c) RT treatment alone.

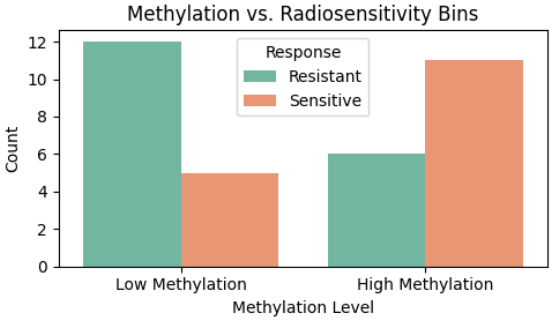
tance. As shown in Fig. 3.2b, a scatter plot of AUC versus mean methylation does not reveal a strong separation between resistant and sensitive samples (defined by the median AUC cut-off of 2.68). When binning samples by methylation (high ≥ 0.35 vs. low < 0.35) and radiosensitivity (resistant ≥ 2.68 vs. sensitive < 2.68), I observe a slight enrichment of resistance among the highly methylated samples (Fig. 3.2).

I examined site-specific samples subsets—lung, ovary, and CNS—selected for their larger sample sizes and greater statistical power. No consistent patterns emerged across sites. Ovary appeared most promising (Fig. 3.3), but a bar-chart comparison showed the difference was not statistically significant.

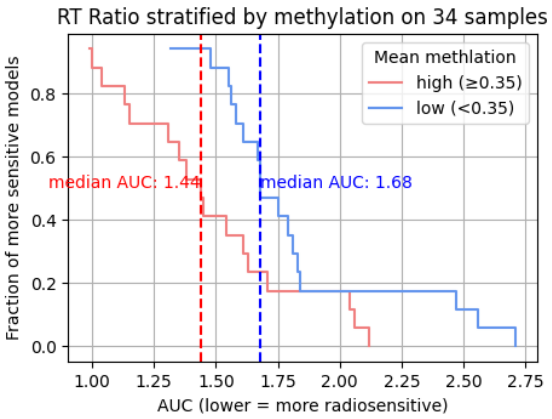
I repeated the analysis in PDX models. Although Kaplan–Meier curves suggested a potential separation, scatterplots showed no statistically significant association (Fig. 3.5).

Methylation of the *MGMT* promoter is a well-established biomarker for response to temozolomide (TMZ) [11]. As a positive control, I confirm that *MGMT* promoter methylation correlates with outcome following TMZ treatment, as well as with combined radiotherapy and TMZ (RT+TMZ), as shown in Figure 3.4.

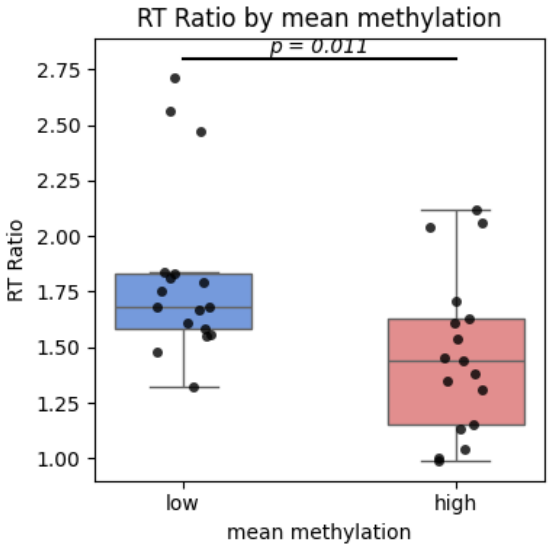
Summary. Global histone modification features show limited standalone predictive value (best $R^2 \approx 0.07$), and global mean methylation is not a robust marker of radiosensitivity in either CCLE or PDX. Motivated by observed missing value issue and high variance, I next test whether imputing missing values can improve relevant signal.



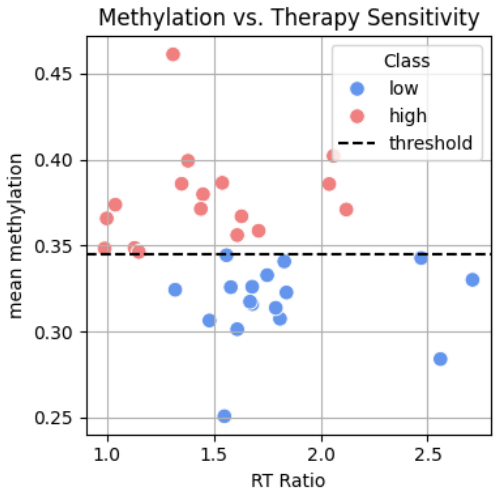
(a) Methylation vs. radiosensitivity bins. Samples are binned by median methylation (0.34) and median RT Ratio (1.6).



(b) Pseudo–Kaplan–Meier plot



(c) RT Ratio by methylation class



(d) Scatterplot mean methylation vs. AUC. Low and high methylation (by median 0.34) groups shown.

Figure 3.5: Global mean methylation and radioresponse on PDX model data. More significant trends observed.

3.2 MOSA imputation boosts RNA-seq and methylation performance but not histone

Aim & approach. To increase number of samples, address missing values problem, and enable more effective multi-omic integration, I consider artificial value imputation. I re-trained MOSA to impute missing values in histone data and then re-benchmarked imputed datasets on classical ML models.

The Multi-omic Synthetic Augmentation (MOSA) model [6] is an encoder–decoder architecture developed to impute missing values and samples in multi-omics datasets. I re-trained MOSA on the six multi-omics datasets used in the original paper [6], replacing the drug-response dataset with global histone modification data. All training parameters followed the original setup, except for the KL divergence loss term. Because of the low dimensionality of histone data, the KL term diverged to infinity; I therefore set its weight to zero. In the original MOSA training curves, the KL contribution was minor, so excluding it is justified.

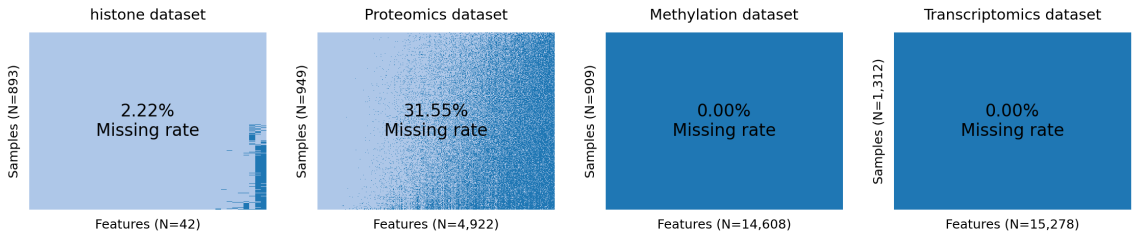
After imputation, the number of histone dataset samples with linked AUC response increased from 440 to 490. MOSA also generated entirely new histone data for cell lines where no measurements existed, but to avoid introducing excessive noise I excluded those and only used imputations to fill existing missing values.

Figure 3.6a shows the missing value rates before MOSA imputation. Even though RNA and methylation appear at 0% missing rate, the samples (rather than only singular entries) will be imputed. Figure 3.6b highlights the lack of overlap between datasets, where sample-level imputations improve integration. Training and validation curves (Fig. 3.6c) confirm stable convergence, with decreasing loss despite high variance across validation folds.

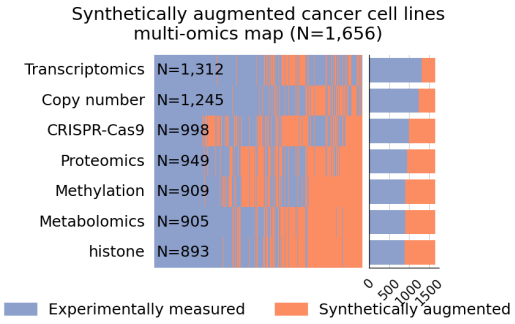
In the following experiments, I use the originally imputed RNA-seq and methylation datasets along with the original histone modification data in which missing values are imputed (no sample imputation). Effect of imputation on performance on classical ML models is presented in Figure 3.7.

In the imputed methylation dataset, the number of labelled samples increases by 47 (498 vs. 451). Despite this modest gain in n, performance improves substantially, suggesting that imputation reduces noise as well as recovering missing entries. For example, Lasso improves from $R^2 = 0.16$ to 0.22 with noticeably lower variance across folds. However, MOSA does not preserve the original methylation scale (beta values in $[0, 1]$), so absolute magnitudes should be interpreted with caution.

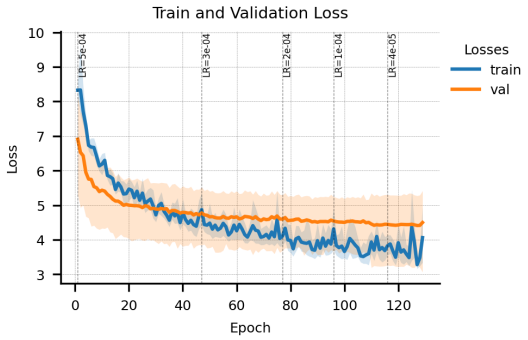
Imputing the histone data adds 50 labeled samples and reduces variance (test



(a) Missing features within the dataset. The percentage of missing values is calculated after removing features and samples with exclusively missing entries. For the methylation and RNA-seq datasets, this filtering step removed all incomplete columns, resulting in no missing values in the remaining data.



(b) Missing cell lines among datasets.



(c) MOSA training and validation loss. Validation shows high variance but a steady decrease; no obvious overfitting.

Figure 3.6: Overview of dataset quality and MOSA training. (a) Missing features across modalities. (b) Missing cell lines across datasets. (c) Training and validation loss of MOSA model.

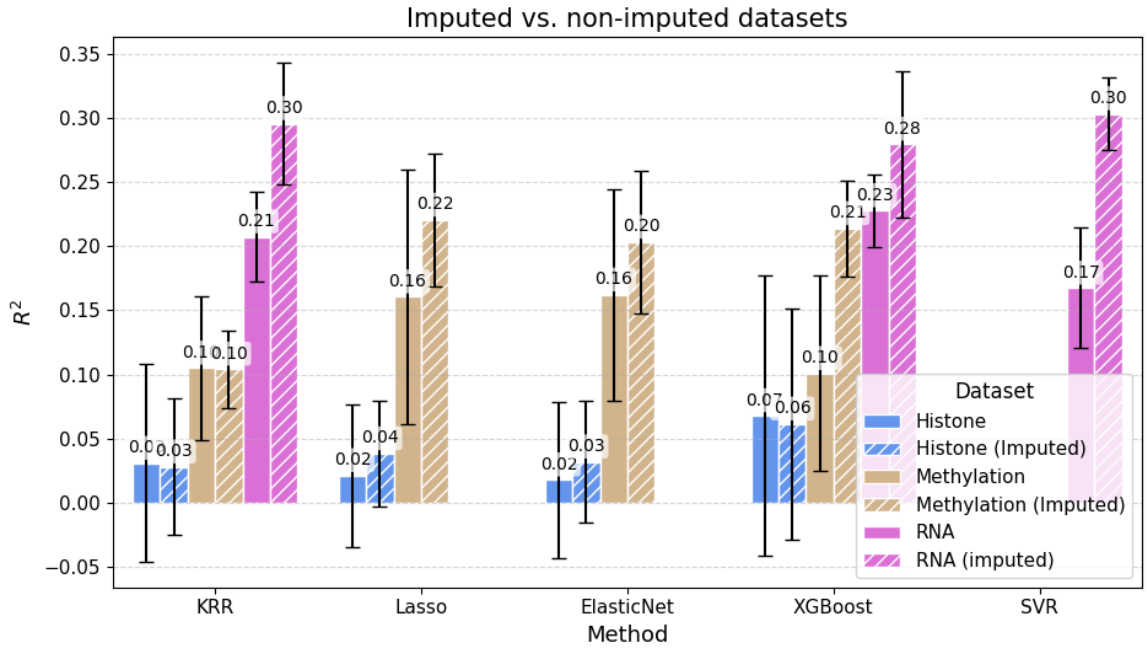


Figure 3.7: Effect of imputed datasets on R^2 metric. Imputation drastically improves performance of both Methylation and RNA-seq. The effect on histone is insignificant.

R^2 variance is reduced from 0.4 to 0.2), but overall predictive accuracy worsens. Therefore, in subsequent experiments I retain the original (non-imputed) histone dataset.

RNA-Seq imputations yield the clearest benefit. All methods improve, with both KRR and SVR reaching $R^2 \approx 0.30$, and SVR additionally showing reduced variance. This gain likely comes partly from the increased sample size, but also from imputation smoothing out noise in the expression profiles.

Summary. MOSA imputation increases labelled sample counts and improves predictive performance for methylation (to $R^2 \approx 0.22$) and RNA-seq (to $R^2 \approx 0.30$), while offering no benefit—and a small degradation—for histone; accordingly, I retain the original (non-imputed) histone dataset in subsequent analyses. Motivated by the remaining small-n/high-d setting, I next evaluate biology-informed dimensionality reduction via GSVA to stabilize training without sacrificing accuracy.

3.3 Pathway aggregation (GSVA on co-essential modules) reduces dimensionality without sacrificing accuracy

Aim & approach. To address high dimensionality with limited sample size, I mapped gene-level RNA-seq, methylation, and proteomics to module activity scores (GSVA) and compared to gene-level baselines.

I use the list of modules from [50] (Supplementary Data 2), which contains 5225 gene sets with enrichment values and p-values. Gene names were standardized and matched to those in the aggregated feature datasets (RNA-Seq, methylation, and proteomics).

Modules were filtered to retain only genes present in the dataset, keeping only sets with at least two mapped genes. I considered three settings: (i) all modules, (ii) "most-enriched" modules ($p < 0.05$, enrichment > 100, as in [50]), and (iii) the top 500 modules ranked by enrichment. For each setting, the module–gene sets and expression data were processed using the GSVA algorithm. The resulting datasets contain the same cell lines as the originals, but features are now pathway-level module scores.

I then trained classical ML models on these reduced datasets, comparing against the original gene-level representations (Tables 3.2, 3.3, 3.4 'Original genes' results were provided by fellow students and are included for comparison). Results on RNA-Seq (imputed) show that module aggregation does not improve R^2, though performance remains in a steady 0.25–0.30 range even with a $\sim$30-fold reduction in features. Interestingly, the 2-layer NN benefits from reduced dimensionality, improving from negative R^2 at the gene level to positive values with module features.

For proteomics, performance differences across KRR, RBF-KRR, and ElasticNet are small. SVR performance decreases as the number of modules decreases, while the MLP again improves with dimensionality reduction. XGBoost trained on the 500-module representation achieves the best overall performance, reaching $R^2 = 0.281$ and outperforming the raw gene representation. Methylation (imputed) followed a similar pattern (Table 3.4), however now the performance was more significantly reduced.

Interpretability of models using modules It's important to note that using modules does not necessarily restrict interpretability. For example, the RNA-seq (Imputed, 500 modules) when trained with XGBoost, I can reveal the most important

Table 3.2: **RNA-Seq Imputed** performance when gene-level features are grouped into modules. Values are mean ± s.d. across folds.

	Original genes	**All modules**	**Most-enriched modules**	**500 modules**
Num. Features	15,278	5,092	1,227	500
KRR	**0.305** ± 0.051	0.2977 ± 0.054	0.2823 ± 0.055	**0.2913** ± 0.043
KRR (RBF)	—	0.293 ± 0.064	0.271 ± 0.051	0.2844 ± 0.0406
ElasticNet	—	0.295 ± 0.056	0.287 ± 0.050	0.287 ± 0.0419
XGBoost	0.279 ± 0.057	—	—	0.253 ± 0.045
SVR	0.303 ± 0.028	0.265 ± 0.072	—	0.287 ± 0.029
2-layer NN	−0.126 ± 0.77	0.215 ± 0, 077	0.207 ± 0, 051	0, 258 ± 0, 046

Table 3.3: **Proteomics Imputed** performance when gene-level features are grouped into modules. Values are mean ± s.d. across folds.

	Original genes	**All modules**	**Most-enriched modules**	**500 modules**
Num. Features	4,922	2,760	977	500
KRR	0.254 ± 0.030	0.244 ± 0.069	0.259 ± 0.037	0.258 ± 0.031
KRR (RBF)	—	0.253 ± 0.063	0.245 ± 0.036	0.246 ± 0.038
ElasticNet	0.263 ± 0.066	0.240 ± 0.056	0.254 ± 0.041	0.254 ± 0.040
XGBoost	0.219 ± 0.050	—	—	**0.281** ± 0.034
SVR	0.257 ± 0.036	0.228 ± 0.55	0.208 ± 0.027	0.204 ± 0.26
2-layer NN	0.150 ± 0.117	0.166 ± 0.120	0.167 ± 0.112	0.215 ± 0.071

Table 3.4: **Methylation Imputed** performance when gene-level features are grouped into modules. Values are mean ± s.d. across folds.

	Original genes	**All modules**	**Most-enriched modules**	**500 modules**
Number of features	17,100	5,000	1,200	500
KRR	**0.259** ± 0.058	0.199 ± 0.086	0.202 ± 0.037	0.196 ± 0.041
KRR (RBF)	0.215 ± 0.039	0.210 ± 0.081	0.192 ± 0.030	0.201 ± 0.036
ElasticNet	0.20 ± 0.055	0.197 ± 0.074	0.20 ± 0.39	0.198 ± 0.040
XGBoost	0.214 ± 0.037	—	—	0.191 ± 0.05
SVR	0.239 ± 0.064	0.158 ± 0.109	0.184 ± 0.017	0.167 ± 0.03
2-layer NN	0.116 ± 0.083	0.096 ± 0.129	0.145 ± 0.092	0.152 ± 0.081

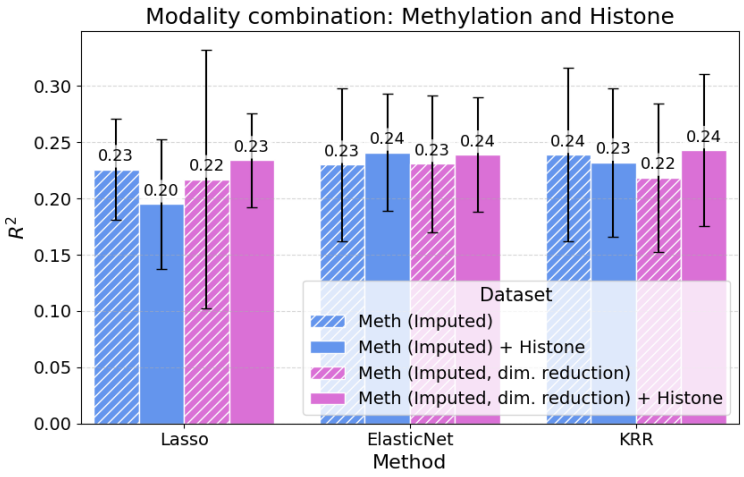
features (using the `feature_importances_` xgboost functionality). They end up being the 4650, 3700 and 583 modules and are known to be related to tetrahydrofolate interconversion, interleukin-1 receptor activity, and Smc5-Smc6 complex respectively. While the proteomics dataset found the 3737 (H3 histone acetyltransferase complex), 2250 (canonical glycolysis) and 1448 (pre-replicative complex assembly involved in nuclear cell cycle DNA replication) modules as most informative.

Alternatives to GSVA for groupig gene modules I have also considered the PLAGE and ssGSEA methods as alternatives to GSVA. Exsisting code can be easily adjusted by using the `gsvaParam` or `ssgseaParam` methods that are part of the GSVA R package. I have calulated the scores for 500 strongest modules only. They proved to perform worse than GSVA on the imputed RNA-Seq data simple ML model benchmarks. With PLAGE reaching best score of 0.16 R^2 on KRR with RBF kernel (while GSVA scores resulted in 0.28 R^2). While ssGSEA scored up to 0.268 (also KRR with RBF kernel), and much lower on remaining models. This result encouraged me to continue using GSVA for the remainder of experiements.
Summary. GSVA compresses features by $\sim$10–30$\times$ while maintaining RNA-seq and proteomics accuracy (RNA $R^2 \approx 0.29$; proteomics XGBoost $R^2 = 0.281$) and only modestly reducing methylation. I then proceed to testing whether fusing modalities can further improve on these results.

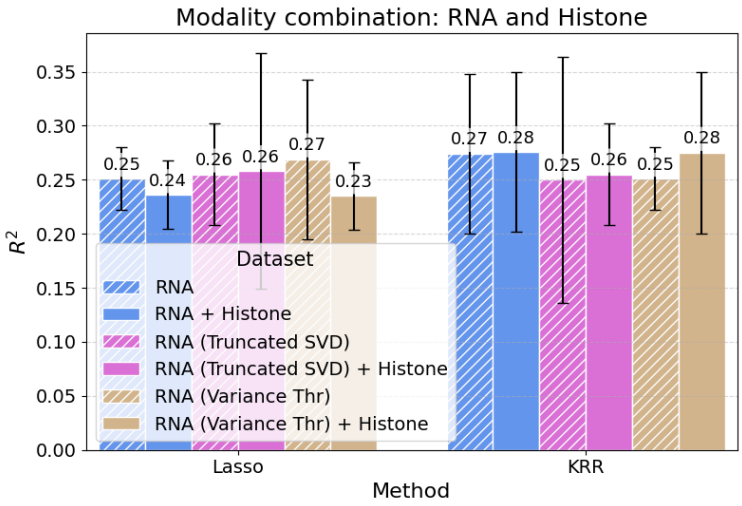
3.4 Multi-omic fusion does not surpass the best single modality; histone often harms

Aim & approach. To test whether multi-omic integration adds complementary signal beyond the strongest single modality, I evaluated three fusion strategies under identical preprocessing and cross-validation: (i) *early fusion* via feature concatenation—both raw and after dimensionality reduction; (ii) *module-level concatenation* using the 500-module GSVA features for RNA-seq, methylation, and proteomics; and (iii) *mid-fusion* with a multi-head MLP comprising modality-specific encoders and a shared joining network.

First, I explore fusion of histone with RNA-Seq and methylation. I use the imputed RNA-Seq and methylation datasets, restricted to the 432 samples with histone measurements. I test two early-fusion strategies for combining histone with each modality: (i) direct concatenation; and (ii) dimensionality reduction of the base



(a) Histone + Methylation



(b) Histone + RNA

Figure 3.8: Early fusion of histone with methylation (a) and RNA-Seq (b). Each panel compares the base modality alone, raw concatenation with histone, and concatenation after dimensionality reduction of the base modality. Bars show mean accuracy with standard deviation across splits. Raw concatenation offers no benefit and can reduce accuracy; after reduction, fusion yields gains for methylation and shows KRR to be more reliable than Lasso when modalities are combined.

modality (e.g., truncated SVD or variance threshold) followed by concatenation. I also train baselines on the base modality alone to isolate the contribution of histone. The results in Fig. 3.8 show that raw concatenation does not exploit histone and often lowers accuracy. After dimensionality reduction, the fused models improve over the single-modality baselines—most clearly for Methylation+Histone in Fig. a. For RNA+Histone (Fig. b), Kernel Ridge Regression (KRR) is more stable than LASSO when modalities are combined. Overall, dimensionality reduction is necessary for effective early fusion; otherwise the high dimensionality of the concatenated representation is detrimental and a single modality may be preferable.

Next, I explore fusion of gene-level data grouped in Modules. To test whether module-based dimensionality reduction improves modality fusion, I use the 500-module RNA, methylation, and proteomics datasets and train classical ML methods on their concatenated features. Multi-omic integration of Histone was excluded due to its consistently detrimental effect on performance. As shown in Table 3.5, concatenation does not improve performance over single-omic baselines (Tables 3.2, 3.3). In fact, dropping one or more modalities can yield stronger results. Across methods, all multi-omic combinations fall within the 0.22–0.26 range, with RNA+Proteomics being the most promising two-modality pairing.

Table 3.5: **Multi-Omics** performance when datasets are concatenated - GSVA modules used (500 modules each). Values are mean ± s.d. across folds.

	Single-Omic (best result)	**RNA+Meth**	**RNA+Prot**	**Meth+Prot**	**RNA+Meth +Prot**
ElasticNet	0.287 (*RNA*)	0.235 ± 0.088	0.254 ± 0.066	0.227 ± 0.072	0.248 ± 0.075
KRR	0.29 (*RNA*)	0.236 ± 0.088	0.253 ± 0.071	0.237 ± 0.078	0.242 ± 0.086
XGBoost	0.281 (*Prot*)	0.252 ± 0.009	0.249 ± 0.093	$\mathbf{0.262} \pm 0.06$	$\mathbf{0.263} \pm 0.085$

Lastly, drawing on the promising MLP results of 500-module RNA, Methylation and Proteomics, I design an MLP with modality-specific heads (Section 2.8). To validate whether the results benefit from fusion, or rather just the architecture, I also train single-head ablation. The training setup is identical: the modality-specific networks use the same search ranges for depth and width, while the fusion block is replaced by a single linear layer that outputs the prediction. I evaluate all pairwise combinations of proteomics, methylation, and RNA-Seq, the full three-way fusion, and configurations that incorporate histone features (Table 3.6). Across pairs, performance lies between 0.215 and 0.255 R^2, with no statistically clear winner. The three-modality fusion reaches 0.244 R^2, but does not surpass RNA-Seq alone (0.266 R^2). Notably, fusing

Table 3.6: **Modality fusion using MLP with modality-specific heads** All of the datasets are the 500-module imputed datasets. RNA alone performs best. I can see that fusion of methylation and proteomics improves the result over using a single modality.

Modality	**Multi-head MLP**
RNA	$\mathbf{0.266} \pm 0.070$
Meth	0.146 ± 0.046
Prot	0.209 ± 0.057
RNA + Meth	0.215 ± 0.045
RNA + Prot	0.255 ± 0.037
Meth + Prot	$\mathbf{0.228} \pm 0.05$
RNA + Meth + Prot	0.244 ± 0.037
RNA + Histone	0.170 ± 0.041
Meth + Histone	$\mathbf{0.20} \pm 0.055$
Prot + Histone	0.125 ± 0.105
RNA + Meth + Hist	0.18 ± 0.06

methylation with proteomics improves over either modality in isolation, indicating complementary signal. I also observe relatively low variance for RNA+Proteomics and for the three-modality model (e.g., standard deviations ~ 0.037), which is lower than for classical ML baselines (Table 3.5), suggesting greater robustness to variation in train/test splits. By inspecting chosen architecture parameters in the grid search I note: (i) a clear preference for strong weight decay (about 0.2); (ii) wider joining MLPs were preferred (e.g., $(128, 32)$); and (iii) early stopping (fewer epochs) having a regularizing effect.

Adding histone features generally degrades performance, with one exception: Methylation+Histone improves over methylation alone (from 0.146 to 0.20 R^2). The overall decrease could stem from a combination of factors: distributional and scale differences that are not fully compensated by the modality-specific heads, lower signal-to-noise in the histone features, and the increased risk of overfitting from adding weaker predictors.

Summary. Across integration strategies, multi-omic fusion did not surpass the strongest single modality. Raw early fusion with histone often reduced accuracy, while dimensionality reduction was necessary to avoid harm (clearer gains for Methylation+Histone than RNA+Histone; Fig. 3.8). At the module level, concatenating RNA, methylation, and proteomics yielded 0.22–0.26 R^2 across methods, with RNA+Proteomics the most consistent pair yet still below the best single-omic baselines (Table 3.5). Mid-fusion MLPs were more stable (lower s.d.) and showed modest complementarity (e.g., Methylation+Proteomics 0.228 R^2), but RNA alone remained highest (0.266).

Histone generally degraded performance, with one exception (Methylation+Histone 0.20 vs. Methylation 0.146 R^2; Table 3.6). Because fusion delivers no gains, I next shift focus from maximizing global R^2 to better capturing clinically relevant sensitivity *extremes* by re-weighting samples and revisiting evaluation metrics.

3.5 Re-weighting samples improves fit of highly-sensitive and highly-resistant instances

Aim & approach. To prioritize clinically relevant tails under label noise and imbalance, I compared unweighted vs weighted training, alternative objectives/metrics, and classification/probabilistic variants.

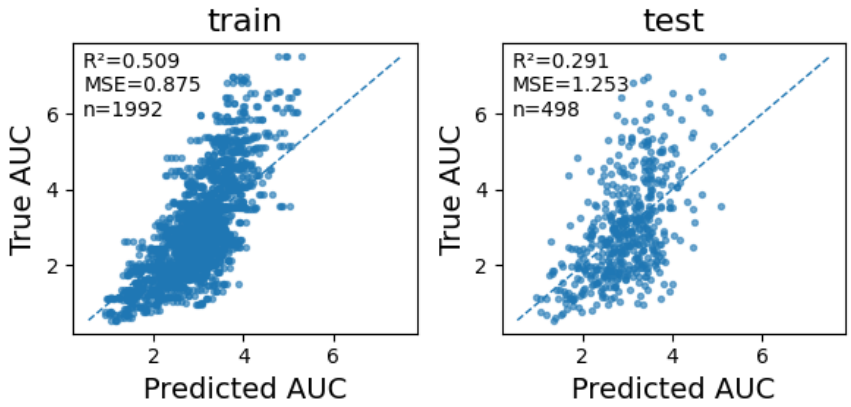


Figure 3.9: Scatter plot of true AUC values vs. predicted ones by the KRR model using RNA-seq 500 module dataset. We observe strong regularization with most of the predictions falling into the $[2, 4]$ range.

The current best-performing model is the 500-module RNA-seq dataset on KRR, achieving $R^2 = 0.29$. Figure 3.9 is a scatterplot of true versus predicted by this model AUC values. It reveals strong regularization, with the model primarily fitting to the *bulk* $[2, 3]$ of the AUC range. While this yields the highest R^2 so far, the model remains poor at distinguishing high from low sensitivity. The model predicts almost all AUC values to be only in the range $[2, 4]$. Given the inherent noisiness of labels and data, this limitation is particularly concerning. To address this phenomenon, we consider three complementary strategies: (i) reformulating the task as a classification problem, (ii) introducing sample weighting such that points farther from the median receive higher contribution, and (iii) employing alternative evaluation metrics beyond R^2 and MSE.

For probabilistic models to address both label and data uncertainty we consider the Gaussian Process Regressor (GPR) and the Bayesian NN. The GPR has achieved at most 0.262 ± 0.058 R^2 across experiments on the RNA 500 module data. After experimenting with combinations of the kernels (`RBF`, `RationalQuadratic`, `WhiteKernel`, `Constant`) and (when `WhiteKernel` not used) varying noise levels (0.05 - 0.3) specified through the α parameter. Best performing models are listed in Table 3.7 and do not yield improvement over the previous experiments.

Table 3.7: **Gaussian Process Regressor** using the RNA-Seq 500 module dataset. Top performing kernel combinations reported; appear to be limited at around 0.26 R^2.

Kernel	R^2
RBF + RationalQuadratic+ $\alpha = 0.1$	0.262 ± 0.058
RBF + WhiteKernel	0.253 ± 0.059
RBF + RationalQuadratic + WhiteKernel	0.252 ± 0.065
RBF + Constant + WhiteKernel	0.251 ± 0.058

I next consider Bayesian Neural Networks. The multi-head MLP was adapted to incorporate the NLL loss and to include dropout for variance prediction. As shown in Table 3.8, this did not improve performance with respect to the R^2 metric. Once again, combining modalities led to a decrease in performance. An advantage of this approach is that the model also outputs predicted sample uncertainty, though this aspect was not explored further given the unsatisfactory results.

I also explored reformulating the problem as classification rather than regression. Specifically, I defined a four-class setup by binning AUC radiosensitivity values into: highly sensitive [0,2), sensitive [2,4), resistant [4,6), and highly resistant [6,8). As this was an exploratory analysis, I report results only for the RNA-Seq 500 module dataset. I evaluated Logistic Regression (LR), a Support Vector Classifier (SVC), and XGBoost, comparing default settings with class-balanced sample weights to address class imbalance. Across models, overall accuracy ranged from 55% to 57%. Performance is shown in Figures 3.10, 3.11, and 3.12. Without weights, LR collapses toward the majority bins (mostly [2,4)), yielding near-zero accuracy in the resistant classes but an overall score around 0.53 due to the skewed distribution. Using balanced weights raises accuracy in the minority classes (e.g., [0,2), [4,6), [6,8)) to roughly 0.3–0.4, at the cost of lower accuracy on [2,4) and a drop in the overall score. Among the three models, XGBoost is the least affected by the class imbalance, though the same pattern remains.

Table 3.8: **Bayesian multi-head MLP** performance on the 500-module datasets and their fusions

Modality	R^2	NLL $\sigma = 0.2$
RNA	0.247 ± 0.070	18.20 ± 2.63
Meth	0.143 ± 0.043	19.3 ± 2.48
Prot	0.204 ± 0.051	18.7 ± 1.68
RNA + Meth	0.184 ± 0.122	
RNA + Prot	0.22 ± 0.101	
Meth + Prot	0.19 ± 0.099	
RNA + Meth + Prot	0.220 ± 0.032	

Table 3.9: **Binary classification** using the RNA-Seq 500 module dataset. Visibly the model still fits to the noise in the bulk instead of distinguishing high and low sensitivity

	Accuracy %	**F1 score**
Logistic Regression	66 ± 5.2	67 ± 5.2
SVC	67 ± 3.1	69 ± 1.2
SVC (RBF kernel)	68 ± 3.2	70 ± 1.3
XGBoost	65 ± 3.3	67 ± 2.7

The binary classification setting (separation at median AUC value 2.7) was also considered, achieving a low 68% accuracy (Table 3.9). This was likely a result of the model still fitting to the bulk region values, learning noise instead of high-low sensitivity signal.

To investigate whether the bulk region is responsible for restricted high from low sensitivity learning, I consider a dataset where the datapoints with AUC values in range $[2, 3]$ are removed. This is a rather exploratory direction, to see whether the models can pick up the true sensitivity signal better when the task becomes easier (fitting highly sensitive and highly resistant samples should be easier than fitting the full spectrum of values). This reduces data from 511 samples to 351. Only RNA on "all" modules is tested and achieves 0.37 ± 0.098 on KRR and 0.36 ± 0.112 on SVR. This reveals that fitting the tail values is possible but perhaps restricted by fitting to the noise of the bulk data. Binary classification on the separated data (classes are "low" for AUC < 2 and "high" for AUC > 3) slightly improves: using RNA all modules achieves $73.4\% \pm 2.6$ on Logistic Regression and $77.4\% \pm 3.5$ on SVM.

These results are not directly comparable with previous explorations since they alter the underlying task. Nevertheless, they highlight that the tails of the distribution carry clearer signal, and that standard training may underweight these informative

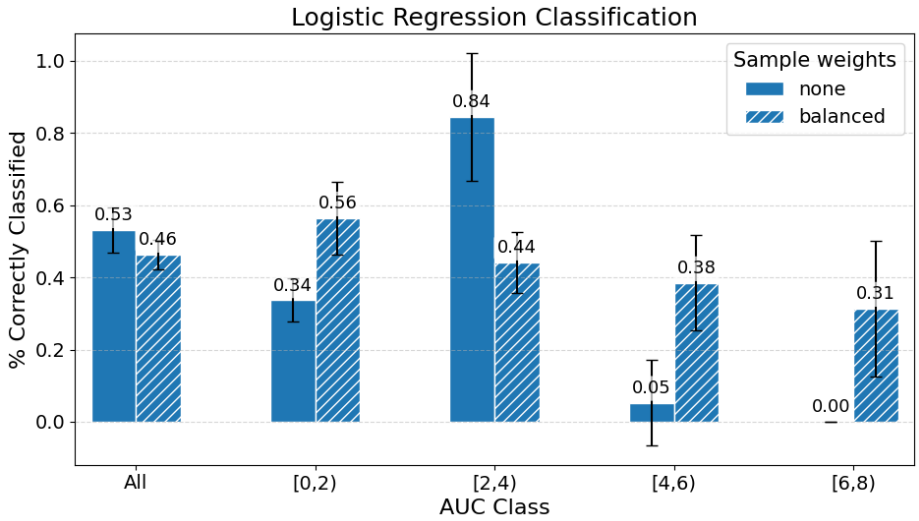


Figure 3.10: **Logistic regression accuracy by AUC class, with and without class-balanced weights.** Bars show mean accuracy for each class and for all samples ("All"); error bars denote standard deviation across splits. Balancing improves the minority classes ([0,2), [4,6), [6,8)) but reduces performance on the majority class ([2,4)) and lowers the overall score.

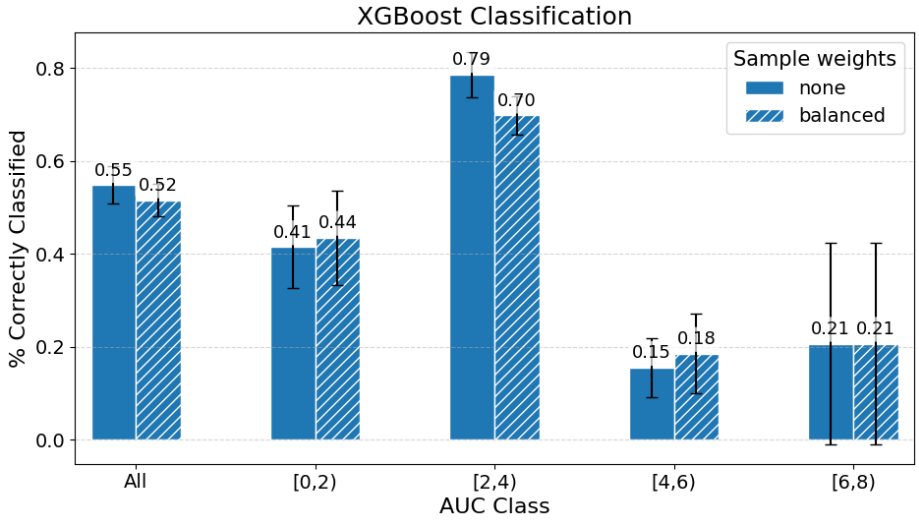


Figure 3.11: **XGBoost accuracy by AUC class, with and without class-balanced weights.** Bars show mean accuracy for each class and for all samples ("All"); error bars denote standard deviation across splits. Balancing only slightly affects the performance.

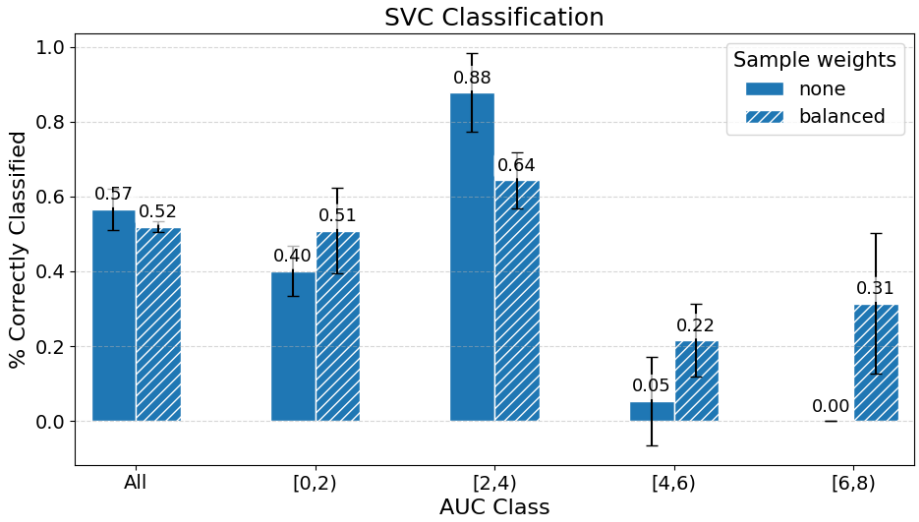


Figure 3.12: **Support Vector Classifier accuracy by AUC class, with and without class-balanced weights.** Bars show mean accuracy for each class and for all samples ("All"); error bars denote standard deviation across splits. Balancing improves the minority classes ([0,2), [4,6), [6,8)) but reduces performance on the majority class ([2,4)) and lowers the overall score.

extremes. A natural next step is to incorporate sample weighting strategy, so that tail samples contribute more strongly to the loss function and the model can better capture clinically relevant sensitivity patterns.

I concentrate on the strong RNA Seq (500 modules) dataset, the KRR model, and consider how sample-weighting strategies influence performance. I use both the MAD-scaled (median-distance based) and Perc-scaled (percentile-based) weights with varying strength of parameter $\alpha \in \{1.5, 2\}$. As shown in Table 3.10, incorporating weights does not improve R^2/MSE performance. However, scatter plots of true vs. predicted AUC from 5-fold CV (Figure 3.13) suggest a more reliable fit. For example, when comparing the best-performing non-weighted model (KRR, $R^2 = 0.29$) with its sample-weighted counterpart (MAD-scaled, $\alpha = 2$, $R^2 = 0.12$), the weighted model clearly predicts a wider predictions range, while the non-weighted version is highly regularized and constrained to [2,4]. Weighting also improves R^2/MSE within training folds by reducing over fitting to noise in the bulk of the distribution.

I next consider alternative evaluation functions: negative log-likelihood (NLL, $\sigma = 0.3$), weighted MSE (MAD/Perc weights with $\alpha = 2$), and an ϵ-intensive metric (ignoring errors within $\epsilon = 1$). These do not consistently align with visual intuition: for NLL and ϵ-intensive metrics, the non-weighted model remains best. Weighted MSE favors weighted models slightly, but not significantly. Rank-based metrics,

Table 3.10: Sample weighting effect on different evaluation metrics. We consider R^2, weighted MSE (using both MAD and Perc weights $\alpha = 2$), mean ϵ-intensive loss ($\epsilon = 1$), Negative Log Likelihood (NLL, $\sigma = 0.3$) and two rank-based metrics Sperman's ρ coefficient and AUROC (class divided by AUC median 2.7).

	R^2	MAD-MSE	Perc-MSE	ϵ-Intensive	NLL	Sperman's ρ	AUROC
None	**0.29**	3.20	1.97	**0.22**	**6.67**	**0.52**	**0.74**
MAD, $\alpha = 2$	0.12	**2.54**	**1.91**	0.31	8.34	0.50	0.72
MAD, $\alpha = 1.5$	0.05	3.17	2.32	0.38	8.91	0.50	0.73
Perc, $\alpha = 2$	0.19	3.15	2.11	0.27	7.57	0.48	0.72
Perc, $\alpha = 1.5$	0.21	3.24	2.10	0.26	7.38	0.49	0.73
Weighted-R^2 used in CV in addition to sample weighting:							
MAD, $\alpha = 2$	-0.25	**2.41**	2.39	0.44	11.7	0.47	0.72
MAD, $\alpha = 1.5$	-0.3	2.46	2.14	0.35	9.7	0.48	0.72
Perc, $\alpha = 2$	0.13	2.69	1.95	0.29	8.14	0.50	0.73
Perc, $\alpha = 1.5$	0.18	2.67	**1.91**	0.27	7.70	0.51	**0.74**

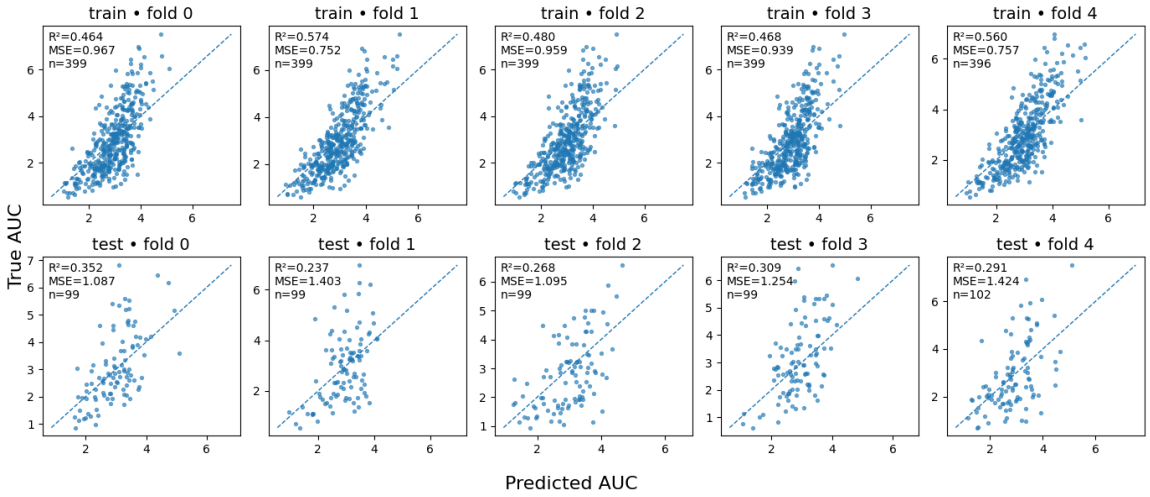
which measure relative ordering rather than absolute fit, again show non-weighted models performing best (Spearman's $\rho = 0.52$, AUROC $= 0.74$), suggesting that even heavily regularized models retain the ability to separate highly sensitive from resistant samples.

Up to this point, cross-validation model selection was based on R^2 for consistency. When I instead used weighted R^2 as the CV metric (with the same weights as in the model) (Table 3.10, bottom half), overall R^2 dropped further, and only modest gains were observed in weighted MSE.

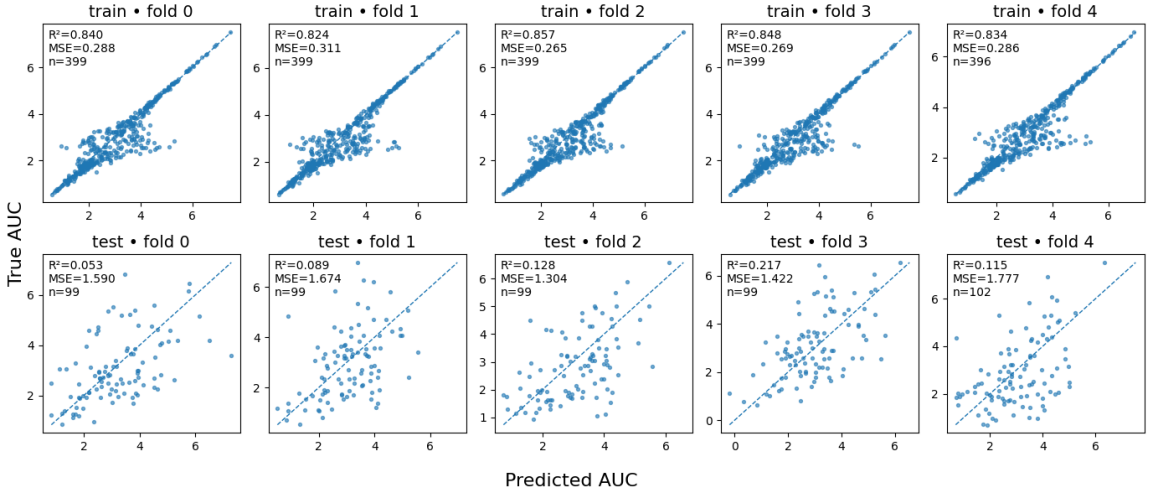
Summary. Standard R^2/MSE push models to fit the bulk (AUC $\sim$2–4) and miss clinically relevant extremes. Sample weighting broadened the predictive range and improved test-set fit visually, while probabilistic and classification re-framings yielded limited gains; tail-only analyses confirmed that the strongest signal lies in the extremes. Finally, I tested whether models trained on cell lines transfer to a more realistic in vivo setting.

3.6 Cell-line models transfer only partially to PDX, retaining modest rank signal

Aim & approach. We want to move beyond cell lines to predict and interpret radiation sensitivity in the in vivo setting of patient-derived xenograft models. We test whether



(a) no sample weights, 0.29 R^2



(b) sample weights, 0.12 R^2

Figure 3.13: Plot of predicted vs. true AUC values accross all train and test folds. Both are the KRR models on the RNA-Seq (500 modules) data. The weighted version used MAD-scaling with $\alpha = 2$. The second plot has better looking test predictions (that span a wider range of AUC values) but much lower R^2.

models trained before translate to the PDX models using scatter plots and rank-based metrics.

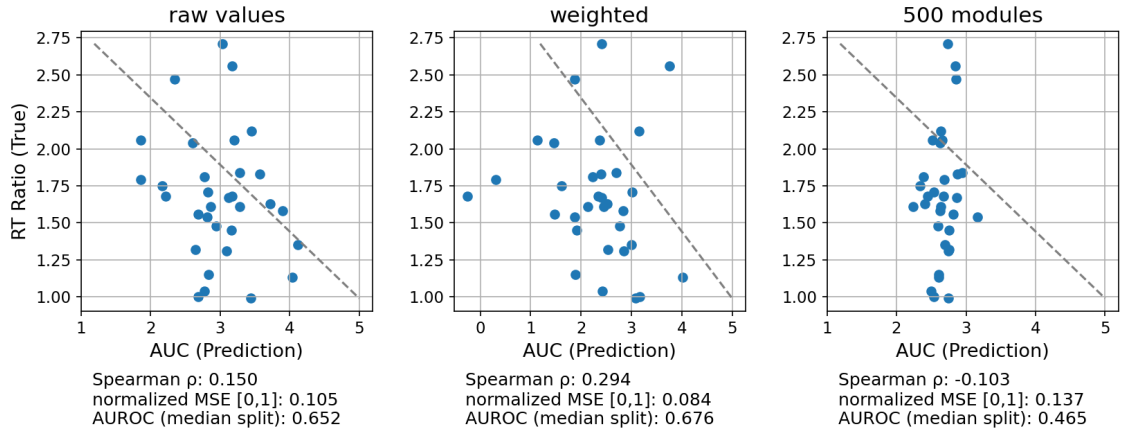
As a reference for expected predictions, glioblastoma cell lines in RadioGX dataset span AUC values from 1.19 to 5.0 (median 2.68, mean 2.83). Filtering by *Histology: glioma* and *Subhistology: astrocytoma grade IV* yields 18 samples, of which 3 are highly radiosensitive (AUC < 2) and 2 are highly resistant (AUC > 4). Thus, reasonable transferred predictions should cover this range, with most samples in the mid-spectrum and relatively few extremes.

The model is trained on the original (imputed, or 500-module) RNA-seq and methylation data, and then evaluated on the PDX Model data (that has been preprocessed as described in Seciton 2.10). Scatter plots have proven to be the most reliable way of evaluating performance (note that AUC and RT Ratio have different scales and are inversely related to radiosensitivity), but I also report R^2, normalized MSE (both AUC and RT Ratio being first normalized to $[0, 1]$), Sperman ρ (sensitivity rank correlation) and AUROC (sensitive vs. resistant).

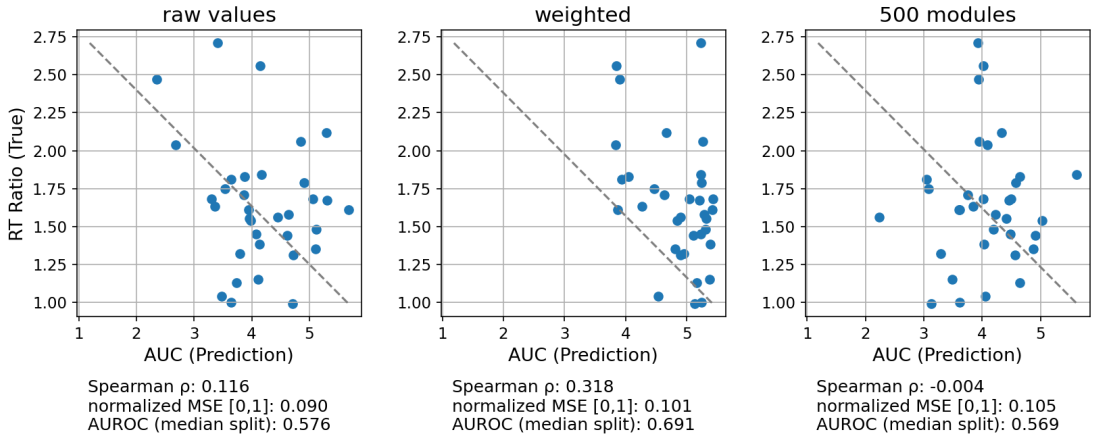
In the scatter plot, the dashed line connects (1.19, 4.31) and (5.0, 1.0), marking the expected extremes for sensitive and resistant glioblastoma samples, respectively. We observe limited transferability: predictions lie within a plausible range and some models follow the anticipated inverse trend. Raw methylation transfers better than RNA-seq on rank-based metrics, likely reflecting more accurate scale alignment during preprocessing. Sample weighting yields small gains (AUROC = 0.69, Spearman's $\rho = 0.31$, normalized MSE = 0.084). Using module features reduces transfer performance, likely due to the heuristic nature of the cross-dataset GSVA transformation.

Exploratory genetic markers. I also tested whether *TP53*/*PTEN* gene mutation status modulates radiotherapy/temozolomide response in glioblastoma PDX; no statistically significant differences were observed (Appendix A, Fig. App. 1).

Summary. Cell line–trained models show only partial transfer to glioblastoma PDX: absolute fit is weak, but rank agreement is modest (Spearman's $\rho = 0.31$) and improves with sample weighting, indicating some retained ability to order low versus high sensitivity.



(a) RNA, KRR



(b) Methylation, Lasso

Figure 3.14: Evaluation of model transferability to Mice PDX Models. Scatterplot of predicted AUC versus RT Ratio (the available radiosensitivity metric for mice models) - note that these are inversely related).

Chapter 4

Discussion

I have explored applying machine learning models for radiosensitivity prediction drawing on the growing available cancer cell molecular data. I have attempted to address the challenges of such data with variational auto encoder data imputation, biology-informed dimensionality reduction and tested its generalizability properties. In particular, I have verified whether histone modification data holds predictive power both alone and when combined with different modalities. Validation on PDX models outlined issues arising when theoretical results are moved to more clinicaly relevant settings.

Raw Global histone modification data is a weak predictor Global histone modification data did not prove to be a strong standalone predictor. The best result was $R^2 = 0.07$ using XGBoost, despite testing a wide range of preprocessing strategies (Figure 3.1). When combined with other modalities, histone features often reduced performance, an effect observed in both classical ML methods (Figure 3.8) and the multi-head MLP (Table 3.6). Applying dimensionality reduction to fused modalities helped mitigate this, allowing models to extract some signal from histone data—particularly when combined with methylation (Figure 3.8a). In multi-head MLP fusion, adding histone to methylation increased performance from $R^2 = 0.146$ to 0.20, though in other cases histone features were detrimental. Overall, the relationship between global histone modifications and radiosensitivity appears complex: in many cases, histone features act more like noise than signal. Previous studies that reported success with global histone markers for radiosensitivity or prognosis typically focused on a small number of carefully selected modifications ($\sim$3–4) [3, 42, 41, 30], rather than the broader panel of $\sim$40 used here. It is therefore likely that many of the available markers are irrelevant in this context and contribute additional noise.

Global Mean Methylation The analyses did not reproduce the strong association between global mean methylation and radiosensitivity reported by Eckhardt et al. [12]. The extended CCLE methylation set and the PDX models failed to replicate the original claim (Figures 3.2, 3.5). Site-specific analyses suggested ovary samples as a possible exception, but results were not statistically significant (Figure 3.3).

Several factors could explain these discrepancies, including differences in the CpG sites assayed, tumor-specific properties, or the model systems used (e.g., PDX models versus cell lines). It is also possible that the original finding does not generalize in a pan-cancer setting—indeed, in ovary cancer the trend was reversed compared with the original study. Even within glioblastoma, where both the mean methylation experiment and the PDX panel had high CpG site overlap (96%), only a slight trend was observed and it was less significant than originally reported.

Overall, global mean methylation does not appear to be a robust predictor of radiosensitivity across either cell line data or PDX models, in contrast to the glioblastoma-focused findings of Eckhardt et al. [12]. By comparison, methylation of the *MGMT* promoter shows a clear association with response to temozolomide treatment (Figure 3.4b), reaffirming its role as a reliable biomarker in glioblastoma [11, 12].

RNA-seq is the best standalone modality Across experiments, RNA-seq consistently emerged as the strongest predictor of radiosensitivity, reaching $R^2 \approx 0.30$ with both SVR and KRR (Fig. 3.7). This aligns with previous studies showing that gene expression is broadly informative for functional phenotypes. Tsherniak et al. [48] found expression to be the most predictive molecular feature of gene dependencies across cancer cell lines, while Seashore-Ludlow et al. [40] similarly identified it as the most reliable predictor of small-molecule response. Together, these results highlight RNA-seq as the most robust single-modality predictor in both my study and the broader literature.

Artificial missing-value imputation improves performance Applying MOSA [6] to impute missing values substantially improved RNA-seq and methylation predictions, with R^2 rising to ~ 0.30 and variance reduced across folds. These gains suggest that imputation not only increases sample size but may also smooth noise, enabling more stable models. In contrast, histone imputations lowered accuracy despite adding samples. The lack of improvement could have been the effect of training setup (switching the KL divergence contribution to the loss), lack of contribution from drug response data (we used histone instead of drug response in training) or the

variable nature of global histone modification data.

At the same time, caution is needed. MOSA was trained on all data, including drug response, which raises the possibility of information leakage and weaker test independence. While imputation proved valuable for downstream modeling, these limitations must be kept in mind when interpreting improvements. It is also important to note that the MOSA imputation does not preserve biologically meaningful values - for example the imputed methylation includes numerous values outside of feasible beta-value [0,1] range.

Gene set aggregation reduces dimensionality but not accuracy I used GSVA [24] to map gene-level data into pathway-level activity scores, based on co-essential modules from [50].

Across RNA-seq and methylation data, module aggregation did not improve predictive accuracy, but performance remained stable ($R^2 = 0.25$–0.30) despite a ∼30-fold reduction in features. Some methods benefited from the lower dimensionality: two-layer NNs improved from negative to positive R^2, and XGBoost on 500 proteomics modules achieved $R^2 = 0.281$—the strongest proteomics result overall. Methylation performance was slightly weaker, likely due to cropping imputed values to the $[0, 1]$ interval. Modules therefore provide an effective dimensionality reduction strategy, improving training efficiency and enabling more expressive models (e.g., MLPs) without sacrificing accuracy.

Subset selection during cross-validation (CV), by contrast, showed high variance and was unreliable. This is consistent with prior reports that in high-dimensional, low-sample multi-omics data, feature selection is often unstable—small perturbations can yield very different gene sets due to correlated predictors. To improve stability and interpretability, biology-informed priors (e.g., pathways or PPI subnetworks) are commonly used [52], and we similarly observed the benefits of using biology-informed feature sets.

The literature also suggests that alternatives to GSVA, such as PLAGE and ssGSVA, may provide greater stability [51], since they compute scores based on standardized expression rather than relative ranks, making them more directly comparable across datasets. However, in our setup these approaches did not yield measurable improvements.

Modality combination Across strategies, naive concatenation proved ineffective, and dimensionality reduction was required for successful fusion with histone data.

Module-based integration did not outperform strong single modalities. Mid-fusion MLPs provided more robust performance and revealed modest complementarity between methylation and proteomics, although RNA remained the dominant predictor (Table 3.6).

I address common challenges in multi-omic integration, including missing values, limited data availability across modalities, and differences in data scale. Such challenges are widely recognized in the literature. For example, Cai et al. [7] note that many multi-omics integration tools do not significantly outperform simple PCA on benchmark datasets, underscoring the difficulty of achieving reliable gains from integration.

Simple ML models often outperform more advanced approaches A key advantage of working with relatively small datasets is the feasibility of applying robust classical machine learning models. For example, kernel ridge regression (KRR) has a computational complexity of $\mathcal{O}(n^3)$ with respect to the number of samples, which would be prohibitive for very large datasets but is entirely manageable here. In practice, classical models often match or even outperform more advanced deep learning methods in the high-dimensional, low-sample regime considered in this study [36]. Increasing model complexity typically introduces a larger number of parameters, which raises the risk of overfitting and can amplify data noise.

Sample weighting reveals limits of classical evaluation metrics Sample weighting shifted model behavior toward distinguishing sensitive from resistant samples, even though standard metrics failed to capture the improvement. Visually, weighted models predicted a broader and more clinically relevant range of AUC values, whereas unweighted models collapsed into the bulk region. This suggests that R^2 and MSE primarily reward fitting the noisy majority, rather than capturing meaningful sensitivity extremes.

Classification offered limited benefit, likely due to ambiguity in where to set cutoffs, while experiments on separated datasets confirmed that the tails contain clearer signal. Together, these findings point to the need for both weighting strategies and alternative evaluation metrics better aligned with the underlying goal—discriminating radiosensitive from resistant phenotypes.

Transferability of models to PDX mice While cell lines provide a tractable system for modelling radiosensitivity, they diverge genetically from human tumours

and often have uncertain properties [49]. This limits translation of models to other datasets, and many promising in vitro findings fail in clinical settings. To address this gap, I evaluated transfering models to glioblastoma PDX, where radiosensitivity is measured as the Radiation Sensitivity Ratio (RT Ratio) (survival with radiation divided by survival with placebo), which is inversely related to the AUC used for training.

Transferability was limited (Fig. 3.14). Direct comparison is complicated by radiosensitivity metric differences (AUC vs. RT Ratio), but both scatter plots and summary metrics indicate only partial alignment: R^2 remains low, whereas AUROC reaches 0.69 and Spearman's $\rho = 0.318$, suggesting modest preservation of rank ordering. Sample weighting during training provided small gains. In contrast, module-based models transferred worse, likely due to cross-dataset GSVA heuristics. Overall this result is possibly due to RNA-scale mismatches from heavy preprocessing, and scale shifts introduced by MOSA imputation. Future work should adopt reference-aware scoring (rsGSVA [46]) to align module kernels to the source dataset. It would be particularly interesting to observe whether grouping genes in modules improves transferability - through gene aggregation reducing gene-level noise and representing more global patterns.

Overall, although cross-system transfer remains challenging, these results indicate partial knowledge transfer from cell line–trained models to PDX, highlighting both promise and current limitations.

4.1 Limitations

This work has several important limitations that should be considered when interpreting the results.

The datasets remain relatively small in the context of machine learning, with the number of molecular features exceeding the number of samples by at least an order of magnitude. This high-dimensional, low-sample regime is common in biomedical applications, particularly when patient-derived data is involved, but it increases the risk of overfitting and instability in feature selection. While large-scale genomic resources continue to expand, corresponding machine learning architectures and appropriately sized training cohorts are still limited.

The use of CCLE cell line data introduces challenges of biological realism. Dish-grown cancer cells provide a standardized and scalable system for analysis, but they do not fully replicate the tumor microenvironment or the evolution dynamics of patient

tumors. Cell lines can accumulate genetic drift over time, further diverging from the tumors they originated from [4]. For this reason, testing transferability to more clinically relevant systems, such as patient-derived xenografts, is essential.

The radiation response labels themselves contain noise [53, 2]. The AUC values estimated from dose–response curves exhibit variability across replicates, and the measured radiosensitivity of cultured cell lines may not perfectly reflect in vivo behavior. The use of temporally aligned CCLE 2019 profiles was intended to reduce discrepancies, but label noise remains a significant source of uncertainty.

There is potential for data leakage when using feature modules or imputation strategies. In particular, pathway aggregation methods and imputation algorithms may inadvertently introduce information from the test set if not performed strictly within each cross-validation fold, which can artificially inflate reported performance. In this study, these steps were not implemented within the CV procedure, representing a limitation.

Model evaluation remains challenging. Conventional regression metrics such as R^2 and MSE may be misleading when predictions cluster around the mean, while classification-based metrics require thresholds that are difficult to define objectively. Alternative evaluation approaches, including uncertainty estimation and weighted metrics, were explored but should be further expanded in future work.

The focus here was on pan-cancer prediction (i.e. comprehensive multi-cancer approach), while many existing studies concentrate on a single cancer type (e.g., lung or prostate) where disease-specific biology may be easier to capture. Expecting a single model to generalize across diverse cancer types is ambitious, yet restricting analyses to a single tumor type would dramatically reduce dataset size and statistical power.

Finally, there are broader limitations associated with applying machine learning in biology. While deep learning and other advanced methods can achieve strong experimental performance, their limited interpretability and transparency remain barriers to clinical translation. As emphasized by Wysocka et al. [52], biologically informed and explainable models are critical for acceptance in biomedical contexts.

Together, these limitations underscore the importance of cautious interpretation of the reported results and motivate directions for future work, including improved evaluation frameworks, and models explicitly designed for biological interpretability and clinical transferability.

4.2 Further Work

Several directions could be pursued to build on this work.

First, the development of a more reliable evaluation framework is essential. Conventional regression metrics such as R^2 and MSE do not fully capture the structure of the prediction problem, particularly in the presence of noisy labels. Future work should focus on designing evaluation functions that better reflect the underlying biology and decision-making context, and then re-assess models against these improved metrics.

The construction of pathway-based modules could be improved. In particular, rs-GSVA [46] allows kernels to be computed with respect to a reference dataset, enabling more consistent module generation within cross-validation folds and better transferability to external datasets such as PDX models. Technical difficulties prevented the use of this approach here, but future work could resolve these issues and improve the quality of module features.

Extending the classification approach to a finer-grained setting may help capture radiosensitivity more realistically. For example, discretizing AUC into multiple bins (e.g., 8 bins across the [0,8] interval) would allow models to distinguish between resistant and sensitive samples at different levels of granularity. Accuracy could then be defined more flexibly, treating predictions as correct if they fall into the true bin or its immediate neighbors—thereby accommodating label noise.

Graph representation learning represents a promising avenue. Graph neural networks (GNNs) have become increasingly prevalent in genomic machine learning [52, 45], offering natural frameworks for multi-omic integration and for leveraging known biological dependencies. Applying these methods to global histone modification data could be particularly novel, building on prior studies that used only a limited subset of histone marks [30].

Incorporating additional sources of biological context could improve predictions. Tissue type, for instance, could be included as a feature by ordering tissues by overall radiosensitivity within each cross-validation fold. Likewise, Bayesian multitask kernel learning (MKL) could enable models to share information across related tasks while accounting for tissue- or modality-specific effects.

Finally, future work should explore incorporating sample-specific uncertainties into training. While such uncertainties are not yet available for radiotherapy response, they have been reported for drug response (e.g., CREAMMIST [54]). Using these as a proof-of-concept could demonstrate how uncertainty-aware models not only improve

predictive accuracy but also provide reliable confidence estimates, which are essential for translational applications.

Together, these directions highlight opportunities for both methodological innovation and biological refinement, with the ultimate goal of producing predictive models that are robust, interpretable, and clinically relevant.

4.3 Conclusion

This dissertation set out to explore whether machine learning can predict radiosensitivity from multi-omics data, with a particular focus on global histone modification profiles, their integration with other modalities, and the transferability of cell line–trained models to more realistic PDX systems. The key findings can be summarized as follows:

- **Addressing high dimensionality.** Gene set aggregation through GSVA reduced feature space by ∼30-fold without loss of predictive accuracy, and in some cases improved stability and efficiency. This enabled the use of more expressive models such as MLPs while keeping computation tractable.
- **Predictive value of global histone modification data.** Histone data alone was a weak predictor of radiosensitivity, and often degraded performance when fused with other modalities. Only in narrow settings (e.g., Methylation+Histone in a multi-head MLP) did it contribute positively, suggesting that global marks capture limited, noisy signal.
- **Multi-modal integration.** Fusion strategies revealed modest complementarity between methylation and proteomics, but RNA-seq remained the dominant modality throughout. Naïve concatenation was ineffective; dimensionality reduction or mid-fusion architectures were required to stabilize training.
- **Addressing noise and imbalance.** Imputation with MOSA improved RNA-seq and methylation performance, though caveats about potential leakage remain. Sample weighting highlighted that classical metrics such as R^2 and MSE reward fitting the noisy majority rather than distinguishing extremes. Alternative evaluation functions (e.g., AUROC, rank correlation) better capture clinically relevant performance.
- **Transferability.** Models trained on cell lines showed only partial generalization to PDX mice data, with modest AUROC (0.69) and Spearman's ρ (0.318).

While this suggests some knowledge transfer, differences in metrics, distributions, and biological context restrict applicability. More robust transfer strategies and reference-aware module methods (e.g. rsGSVA) are needed.

In summary, the study shows that while RNA-seq remains the strongest standalone predictor of radiosensitivity, dimensionality reduction, careful imputation, and tailored evaluation strategies can improve robustness. Global histone modification data, at least in its current form, adds little predictive value. Finally, transferring predictive models from in vitro to in vivo remains a major challenge, underscoring the importance of developing both better model systems and methods that explicitly account for noise, imbalance, and domain shift.

Code Availability

The implementation of the described architectures and experiments is documented and made available https://github.com/anonymous36465/UCL_COMP0158.git.

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Appendix

A TP53 and PTEN Gene Mutations — Mice PDX Models

I tested whether *TP53* and *PTEN* mutation status modulates response to radiotherapy and/or temozolomide in PDX models. Using the mutation data, each PDX line was labeled as mutant (mut) or wild-type (wt) for *TP53* and *PTEN*, then merged with therapy response measures (RT Ratio, TMZ Ratio, RT&TMZ Ratio; ratios > 1 indicate longer survival under treatment). Models were grouped into four categories—TP53 mut / PTEN mut, TP53 mut / PTEN wt, TP53 wt / PTEN mut, and TP53 wt / PTEN wt—and the distributions of survival ratios were visualized with boxplots overlaid with jittered points (Figure 1). No statistically significant differences were observed across groups, contrary to our expectation.

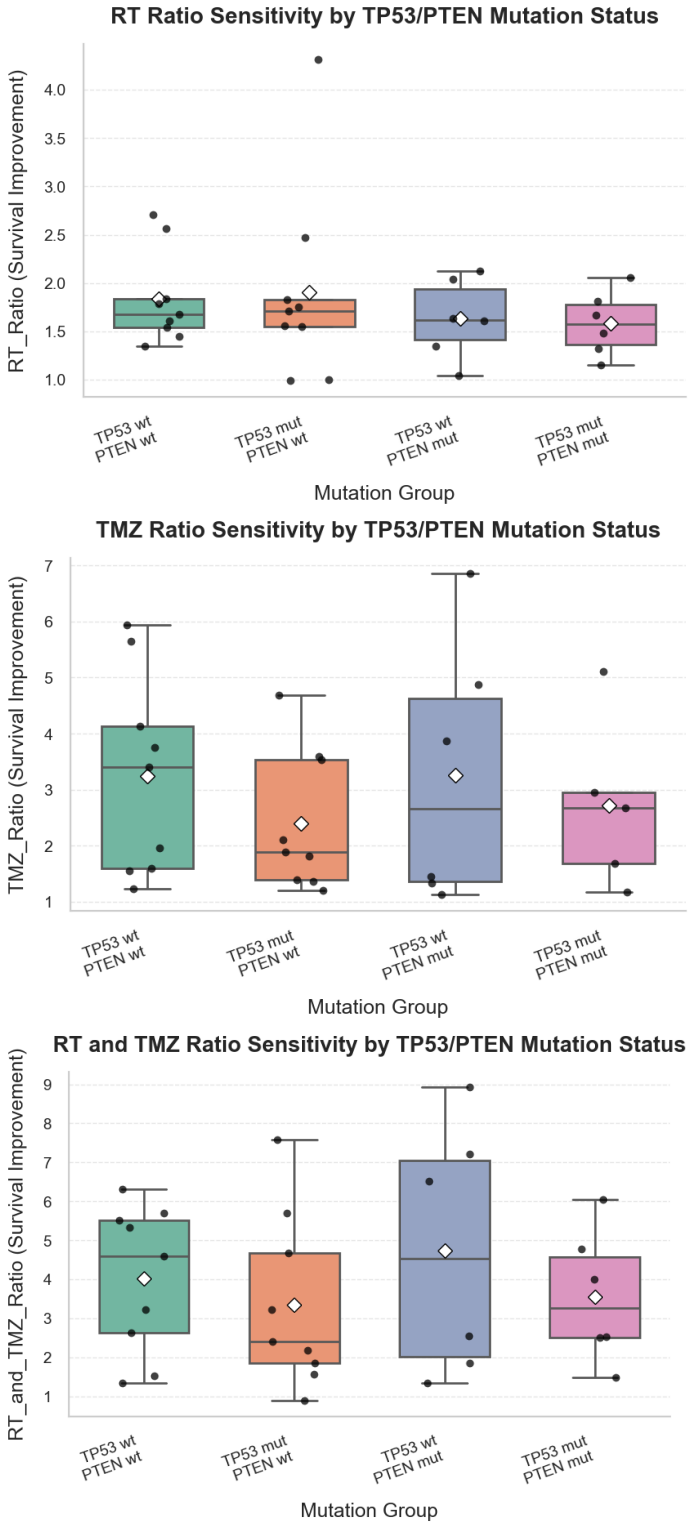


Figure 1: Association of the TP53 and PTEN gene mutations with the ratiotherapy (RT) and timozone (TMZ) response in PDX models. Suprisingly, no association observed.

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